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Energy Management Systems Review

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Summary:

With the recent energy instability in Europe and growing concern for global warming in general, there is not only an interest but rather a necessity for society to conserve energy. Energy Management System is a software tool designed to make building owners aware of energy consumption and control functionality to save energy. There is a plethora of energy management systems available on the market, and this paper aims to review many of these systems as well as relevant papers, regulations and standards governing these systems, and to find what functionality is currently present in the market and how they compare.

Preface

This is a project report on the topic of “Energy Management System Review”. In this report, we have tried to present all the important and necessary information regarding the topic.

The report discusses mainly the standards and regulations regarding the building management system. Different building energy management system vendors and their product functionality has been reviewed. Besides a product comparison has been discussed for better understanding.

We are thankful to our supervisor Professor Carlos F. Pfeiffer. He encouraged us to work on this topic and guided us with necessary information and advice which helped us to complete this project on time. Working on this topic we gathered a lot of knowledge about the BEMS industry. We believe we can implement them in the professional field.

Porsgrunn, <18th November 2022>

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Nomenclature

BEMS: The building energy management system is often referred to as BMS/BEMS.

EMIS: The analysis and monitoring system for energy is known as Energy Management Information System (EMIS).

EnMS (EMS): EnMS is a management system that empowers organizations to plan, review, improve, and identify energy consumption patterns in an organization.

IoT Internet of Things

ISO 50001:2018 is the latest published international standard for Energy Management systems (EnMS)

HVAC: A term to refer to Heating, Ventilation, and Air Conditioning.

1. Introduction

With the increase in the population, energy consumption is rising day by day. Fossil fuel still leads or the production of electricity that is being consumed worldwide. Unlike renewable sources like wind, solar or hydro, fossil fuels have negative impacts on the environment which results in greenhouse gas emissions and global warming, etc. To tackle these problems people are not only focusing on alternative, clean energy sources but also putting emphasis on awareness regarding more efficient use of energy. Power plants, residential and nonresidential buildings are now implementing different kinds of strategies to reduce environmental hazards and minimize energy waste. The conventional electrical grid has been transformed in recent years into an intelligent, very dependable, completely automated infrastructure, preparing the way for the so-called "smart" grid paradigm.

On the demand side, buildings now include digital sensing and communication equipment, enabling continuous energy usage monitoring, intelligent appliance management, grid and utility connectivity making them versatile and working as a smart home. Over the last few years Building Energy Management system (BEMS) has grown a lot of attention. BEMS is an energy management process of using energy more efficiently and helps reduce cost, consumption, and the maintenance of the building equipment. In order to aid in forward planning and cost, BEMS may therefore enhance management information via trend tracking performance. The employees may become more conscious of energy efficiency as a result of this.

This paper provides brief reviews of literatures studies on BEM. Provides more information regarding different strategies and case studies implemented on BMS. It sheds light on EU rules and regulations for building and regulations for Norway. Also discusses ISO 50001 standard and its key elements.

The remainder of the paper is organized as follows. In section 4 different protocols and network system for BEMS have been defined. Section 5 presents different consumer software and their functionalities. Section 6 introduces different BEMS providers and gives an overview of their products. Section 7 summarizes the report. [1][2][3]

2. Relevant Papers

This section covers papers relevant to energy management design and performance.

2.1. Analysis of BEMS

A paper written in 2006 at the National Technical University of Athens named “Intelligent building energy management system using rule sets” looks into the development of a building energy environment system from scratch, targeted towards a typical office building in Athens, Greece. They have produced a system based on a back-end program configured to collect data from multiple different sensors both inside and outside a building, and do rule based decisions in order to optimize power consumption, primarily through managing air-conditioning units and light levels. [4]

The data input devices used in this project us a collection of multiple sensors mounted in each room of the building in addition to hallways and common areas and even outside the building. The sensor types included is measuring temperature, humidity, CO₂ concentration, luminosity, and movement. The data is collected and stored in an MS Access database.

A simple graphical user interface to set up the system with target ranges is given, but aside from that there isn’t anything to view historic data, performance metrics or current operation status. Neither does it do any sort of prediction or reporting. In this aspect it lacks much of the functionality present in retail energy management systems that is required by ISO 50001.

The control algorithm used to act on the collected data is an expert system build on C Language Integrated Production System, a Nasa developed programming language used to emulate an expert making decisions based on the input data by primarily running a series of If-Then rules. In other words, this system does not control its actuators based on a black box neural network, mathematical model or predictions, but rather a predefined static set of checks and rules.[5]

This system was tested on a 485m² office building in Athens over a 12-month period and recorded a 10% decrease in total energy consumption of the facility, while reliably maintaining the set levels for temperature and luminance and general air quality. It’s also stated that most of the energy savings came from avoiding running HVAC equipment and lighting in unoccupied spaces.

2.2. Preliminary case study on BEMS

This case study from University of Applied Science Western Switzerland is a preliminary study on BEMS for one of the universities campus buildings looking to develop a BEMS. This study does not look at the technical aspect of the BEMS development, but rather the needs and desires of the different stakeholders in the project, from upper managerial to users and maintenance staff. [6]

The methodology used is thorough and goes through multiple analyses steps. The first step they outline is identification of stakeholders where they identify who has any interest or impact on the energy management system. They then do a pre-analysis of the stakeholders and rank their influence on the project. When this is done, they move on to face-to-face interviews where they map the stakeholders “motivations, misgivings, needs and expectations”. Lastly, they do a workshop with the stakeholders and a prototype of the system to identify any further needs.

The findings of this study is in ways tied into the specific case they are analyzing, but there is still much to draw from it. They emphasize that the information used for data analyses and automatic decision making is rarely very useful for the users of the system, and that the displayed information should be tailored for the different stakeholders. An example they point out that kilowatt-hours is not a meaningful metric to a person without a sufficient understanding of electricity and especially if presented without any reference points.

Frequency of reports is also highlighted as a point. For directors and accounting a yearly report is enough, while department and facility managers see more benefit in monthly reports, and maintenance personnel would prefer weekly reports. And these reports should also be tailored with the information relevant for the areas each of the recipients manage.

Another important point this paper makes is that there is many stakeholders in the deployment of an energy management system, and that the system should be extended to meet the needs of not only building managers but also aid maintenance. The system should not only help energy efficiency and report on it, but also detect and diagnose problems within the building and sub systems, as this will not only contribute to better indoor environment but also reduce energy performance by speed up repairs and help and reduce the time spent running on faulty hardware or sub-optimal devices.

2.3. EMS paper review

Dasheng Lee and Chin-Chi Cheng from the National Taipei University of Technology have done an EMS paper review, going through 276 papers surrounding the topic of EMS with the aim of finding statistical energy savings as well as categorizing functions and identifying the most important functions for an EMS in regard to energy savings. The papers in the review is written between 1974 and 2014, and thus spans 40 years of research. The authors have categorized these papers into three primary groups; BEMS, industrial/factory EMS, equipment (HVAC/Lighting) energy system. [7]

The cases represented in this review also originate from primarily north America, Europe and Asia, but is relatively balanced between the three areas. It is also implied a balance between hot and cold climates where HVAC system serve different purposes.

The paper outlines energy savings as “ratio of the energy usage difference before/after implementing the EMS”. On an overall view the findings was that BEMS gave an energy saving between 11.39% and 16.22%, and industrial and factory energy management systems had an energy saving of 10.35% to 18.89%. Equipment specific energy savings is broken down into artificial lighting and HVAC. Energy management systems focused on lighting saw the biggest reduction on energy usage with an average saving of 39.5%, while HVAC specific systems reduced on average 14.07%, and the highest reported is 46.9%. EMS systems that included installing renewable energy sources or reported energy savings based on simulations are excluded from overall energy saving findings.

The paper also analyzed the different function of EMS systems and identified 194 separate functions which they grouped into 6 main categories:

Basic function group: The basic function group consists of energy use monitoring functions, reporting and awareness. This can also include calculating energy use index or efficiency indicators. Monitoring functions include the act of collecting the data from sensors. Reporting functions is the production and electronic distribution of periodic reports. Awareness can

include things like a public screen/display with live metrics or indexes on real time performance.

Control function group: This function group encompasses the main control of building actuators, like time scheduled or occupancy based on/off control, indoor thermal and air quality control, balance of energy and comfort control, and time varying energy control based on energy prices or tariffs. For industrial application this function group also include peak load control, power quality control, and demand limit control.

Analysis function group: The main functions of the analysis function group is energy demand prediction and forecasting, model optimization, equipment fault detection and predictive maintenance.

Management function group: This function group manages the control group, as well as organizes staff management. The staff management includes behavioral management in a sense of following up the staff on their energy management related tasks. An example is the ISO 50001 plan, do, check and act routine, requiring staff to follow up on the EMS performance.

Advanced function group: The advanced function group combines the control group and analysis group to get better control based on more advanced mathematical modeling, expert systems, or artificial intelligence.

Specific function group: This function group include control and analyses functions tailored to a specific installation or specific type of equipment. This includes building specific cold/heat storage saved up from off-peak or directing daylight through the indoor space to reduce energy used for lighting.

The findings of this paper is the key functions that contributes the most to energy savings. The biggest contributor is optimized HVAC control with feedforward human interaction. For lighting systems the most contributing function is light intensity control, dimming light sources while keeping indoor lighting at an acceptable level aided by sunlight. For BEMS in general the biggest contributors are scheduled control and tariff/cost control.

2.4. Optimization of Energy Management in Commercial Buildings with Weather Forecasting

This review done by Dimitris Lazos n, Alistair B. Sproul, Merlinde Kay from the university of South Wales goes through different energy forecasting methods, and their application in the energy management of commercial buildings while used in conjunction with load and demand forecasting, energy prices information and different control techniques. [8]

According to their research, commercial buildings pose and advantage for research given their somewhat predictable energy consumption profile, they have facilities managers who are concerned in obtaining energy savings, there are more diverse energy sources in the building besides the grid operator, such as: Solar and Wind energy and own generators. Peak shaving is very significant and preconditioning of the internal environment is possible due to the fact that the buildings are mostly vacant during the night.

The authors go through several research papers describing the techniques used for weather forecasting. They divide them in three types: Statistical (Autoregressive and Linear Regression Models), machine learning methods (Neural Networks and Support Vectors), and physical and numerical (Engineering white and grey methods, Numerical Weather Prediction Methods or

NWP, and physical weather forecasts using external inputs). Their research shows that each of these types has applications and challenges: Statistical methods are simple and fast, with a somewhat acceptable accuracy, however, historical data is needed, and the weighting of coefficient can pose a challenge.

Machine learning methods are accurate, do not need supervision and can describe non-linearities better. On the other hand, they too depend on historical data and are computationally complex.

Regarding the weather variables to predict, it is found by the authors that Ambient temperature, sky modeling and relative humidity are the most related to energy consumption in commercial buildings. Ambient temperature serves as input in the building model for the internal zone temperature prediction. It also affects components like PV modules. The sky modeling helps forecast the incident radiation in flat surfaces, which in turn helps forecast the internal temperature of the building and estimate the performance and output of solar panels where those are installed. The relative humidity is a factor that affects the occupancy comfort and definition of an optimal set point temperature.

In regard to energy generation and given the current pervasiveness of solar installations in commercial buildings, there is value in short-term weather forecasting, this will help predict the peak energy production for the following day, as well as the shading and heat losses.

The authors of this paper found that most commercial buildings favor the use of load forecasting opposed to weather forecasting because this is information that is readily available, obtained from historical data. The load forecasting is achieved mostly through statistical methods and is done for short prediction horizons, to predict the loads of the next following 6 hours for example. The variable that affects load prediction is ambient dry bulb temperature.

Nevertheless, the use of weather data and accurate building models increase the accuracy of the loads forecasting results.

The most important load components in commercial buildings are HVAC and lighting related. With HVAC being the most sensitive to savings in peak prediction and shaving.

One extra tool that new buildings have is the possibility to use the internal thermal energy by storing ice or hot water when there are lower energy prices. This energy saving can then be used when higher prices are expected.

Different control techniques are proposed, which integrate weather data, energy generation and storage capabilities, load forecasting, energy prices, and occupant comfort considerations to optimize the energy consumption and the economical expense. Of the studied techniques, the most interesting is the implementation of model predictive control, with which according to their findings, generates savings of about 50%.

2.5. Building energy-consumption status worldwide and the state-of-the-art technologies for zero-energy buildings during the past decade

This review made by Xiaodong Cao, Xilei Dai, and Junjie Liu from the Tianjin University in 2016, provides a general description of the energy consumption of buildings in the world's three biggest economies: United States, the European Union, and China. It also gives advice in

terms of what is needed to achieve Zero Energy Buildings (ZEB), which are buildings with marginal or zero carbon emissions during a balancing period, which in most cases is a year. [9]

The study shows that for the European countries and the U.S. Buildings take a considerable proportion of the energy demand, with a 40%, whereas in China, this value is considerably less, of about 27%. In these regions, the major building energy consumption goes to space heating: In the U.S. for residential and service sector it accounts for 37% and 25% respectively. In China those values are 71% and 68%, whereas in the E.U. it is 66% for residential and 39% for service sector. These economies still obtain most of their energy from fossil fuels, more so if it is considered that a big proportion of their electricity is generated with fossil fuels.

As buildings represent a big proportion of society's energy consumption, the development of ZEB will help reach the sustainability goals attained in the Paris Agreement. ZEB can achieve marginal levels of consumption by being properly designed, constructed, and operated. The E.U. regulations established a goal of all new buildings to be ZEB in 2020, while in the U.S. this target is of 50% for 2040 and 100% for 2050. In China, when this paper was written, the goal is that 20% of all new construction space be green buildings. The most favored method to assess a ZEB is the balance of energy demand vs. renewable energy generation.

ZEB are achieved through the combination of passive energy saving technologies (Advanced insulation, passive heating or cooling and thermal energy storage), Energy efficient systems (Mostly advanced and efficient HVAC technologies), and renewal energy generation techniques, of which Solar energy and wind energy are the most common.

Building Energy Management Systems (BEMS) are fundamental to increase the efficiency of ZEBs. According to this study, the most important functions used in BEMS are scheduling controls, tariff and loads control. With these functions in place, savings ranging from 11.39% to 16.22% are achieved. BEMS can control different household devices, renewable energy systems, and interact with smart grids using wireless protocols like Zigbee for connection. With these configurations in place, it is possible to control and monitor the performance of the ZEB.

2.6. A review of strategies for building energy management system: Model predictive control, demand side management, optimization, and fault detect & diagnosis

This paper published in 2021 gives a review on energy management system strategies for non-residential and residential buildings to understand what kind of techniques are being used for different kind of buildings for efficient energy use, energy savings and future plan with EMS systems. According to this paper energy consumption by buildings represents 30-35% of global energy use. This large consumption is mainly because of the building sub system such as cooling and heating, lighting etc.

This paper gives an idea about different types buildings that can be designed according to the plan and requirements and categorizes them in 3 major parts. (1) Green Building, (2) Net Zero building, (3) Smart Building. Green building is more on environment friendly side. Net zero building is focused on supplying its own energy by conservation and renewable energy generators. Smart building focuses on occupants comfort and energy consumption.

According to this paper, 2 BEMS approach is considered: 1. Active, 2.Passive.

Active method is concerned with smart approach to reduce energy waste and consumption. Whereas passive approach is concerned with providing future strategies and improving users awareness.

Active method has been categorized in 4 important strategies: (1) Model predictive control, (2) Demand side management, (3) Optimization and fault detection, (4) Diagnosis.

Model Predictive Control: It can predict and forecast buildings response to control requests which is important for better judgement towards decreasing CO₂ emission and energy utilization. For building forecasting 3 approaches are used: White box, Black box, Grey box.

White box method is a straight forward physics based calculation to explain energy performance of buildings. Black box method is data driven, statistical analysis and depends on AI to evaluate energy utilization. Grey box is merge of previous two.

Model predictive control is mainly used in non- residential buildings such as offices and universities. Most researched subsystem in this model is HVAC system.

Demand Side Management: It is a procedure to enhance the energy system on the user side by utilizing improved resources. It has two approaches: Demand response, Energy efficiency. Demand response is applied only to non residential buildings and energy efficiency is applied in both residential and non residential buildings. Most researched subsystem in demand response is HVAC and appliance consumption for energy efficiency.

Optimization: This method has two approaches: Stochastic and robust.

Stochastic approach is focused on occupants comfort index, utilizing minimum power consumption, identifying energy consumption pattern, energy efficiency performance, energy savings etc. This approach is applicable for both residential and non residential buildings and it has been researched for all building systems. Robust approach focuses on optimal planning on local energy system, managing occupants comfort and energy utilization and coordinating cooling systems. This approach is applicable for non residential buildings and most researched subsystem for this approach is HVAC.

Fault Detection and Diagnosis: Fault detection and diagnosis (FDD) is used to identify and isolate BEMS defects in order to protect a system from additional damage. The FDD strategy for the area of BEMS may be divided into two types of techniques: knowledge-based and data-based. Artificial intelligence is used to address FDD difficulties through data-driven techniques. Studies on BEMS that employed a data-driven methodology concentrated on identifying unusual operating patterns and the root cause of heating system failures. Whereas knowledge-driven approaches rely on professionals to identify and discover problems. It focuses on analytic analysis for an air handling unit (AHU). It is also worth noting that both approaches are mainly used in non- residential buildings. HVAC system has been most researched in knowledge based approach and all building system has been highly researched by data driven approach. [10]

From this paper it has been found that 71.74% research efforts are concentrated developing BEMS for non residential buildings, 21.74% on residential buildings and 6.52% for both. 56.52% research focused on HVAC system for non residential buildings. 4.35% research focused on residential HVAC systems. For non residential buildings most used strategy is found to be Model predictive control and for residential buildings is demand side management.

3. Standards and Regulations

Standards and regulations set the baseline requirements for energy management systems. This chapter covers the most relevant standards.

3.1. EU regulation - Energy performance of buildings

The energy performance of a building should be defined based on estimated or actual annual energy used to fulfill various requirements according to its regular use. It should include the amount of energy needed for household hot water needs, heating, and cooling to keep the building's desired temperature. A clear and understandable presentation of the building's energy performance is required, as well as a numeric indicator of primary energy use based on primary energy factors per energy carrier. This indicator should be based on national or regional annual averages or specific value for on-site production. [11]

The methodology for calculating the energy performance of buildings should take into account European standards and shall be in accordance with relevant Union legislation, including Directive 2009/28/EC.

The methodology shall be established with at least the following factors:

- the building's real thermal properties, as well as its interior partitions:
 - (i) thermal capacity;
 - (ii) insulation;
 - (iii) passive heating;
 - (iv) cooling elements; and
 - (v) thermal bridges;
- heating installation and hot water supply, including their insulation characteristics;
- air-conditioning installations;
- natural and mechanical ventilation which may include airtightness;
- built-in lighting installation (mainly in the non-residential sector);
- the design, positioning and orientation of the building, including outdoor climate;
- passive solar systems and solar protection;
- indoor climatic conditions, including the designed indoor climate;
- internal loads.

The positive effects of the following factors shall be considered in the calculation

- local solar exposure conditions, active solar systems and other heating and electricity systems based on energy from renewable sources;
- electricity produced by cogeneration;
- district or block heating and cooling systems;
- natural lighting.

3.2. Norwegian regulations

Norwegian law stipulates regulations on energy declaration of buildings above 50m² in usable area, with extra conditions for corporate buildings above 1000m². This is a required energy declaration that must be executed by professionals and gives the building an energy grade between A to G, where A is the best grade, and G is the lowest, in addition to a list of recommended actions in order to optimize energy consumption. For smaller buildings it is only required to get an energy grade evaluation on sale or when renting to new occupants. For bigger buildings energy grade should always be available and must be renewed every 10 years at minimum or after renovation projects that affect the building energy profile. While the energy grade doesn't come with any other obligation or tax incentives, it is designed to make building owners responsible for their buildings energy performance and spread awareness of energy consumption. [12][13]

An energy management system can assist in meeting this regulatory requirement in several ways. The data collected can contribute to the basis for the energy grading declaration, and if the providers of the energy management system have the required expertise, they can also issue this declaration. The energy grade also gives a potential buyer or renter a quick and easy rating of the building's energy consumption, and with rising energy prices and more awareness of carbon footprint it has the potential to improve sale or renting prices.

The European Unions have directives for the same type of graded certificate in an effort to meet greenhouse gas emission targets, but as of now this is only a voluntary schema for non-residential buildings. A European Commission market study found that buildings opting to use this grading scheme had their prices increase with better energy profiles.[14]

3.3. Iso 50001

Iso 50001 is the international standard governing energy management systems. This standard covers the broad requirements of such a system. [15][16]

3.3.1. Basic definitions

ISO 50001: The latest international standard for Energy Management systems (EnMS) has been published in 2018.

EnMS: EnMS is a management system that provides organizations the possibility to plan, evaluate, enhance, and determine energy consumption characteristics in an organization.

Energy definition: According to the Energy Management System, energy is described as electricity, fuels, steam, heat, compressed air, and other related media. It is significant to note that, for the purposes of Iso 50001, the term "energy" refers to a wide range of energy sources, including renewable energy that can be obtained, stored, treated, used in machinery or a process, or recovered.

Energy Performance: it is described as measurable results related to energy efficiency, energy use and energy consumption. It is vital to remember that energy performance can be evaluated in relation to the organization's goals, energy targets, and other performance standards.

3.3.2. Iso Framework:

- Create and enforce an energy usage optimization policy.
- Describe and develop objectives that adhere to the EnMS policy.
- Explicate a system for the use of data to assist in understanding and making informed decisions about energy consumption.
- Review, monitor and evaluate the results.
- Examine and evaluate the effectiveness of the energy policy.
- increase continuously the reliability of the energy management system as well as energy performance.

3.3.3. Iso Advantage:

- improved operational effectiveness for energy use.
- Reduce energy consumption and negative environmental effects.
- Data organization for fact-based decision-making in the energy sector.
- Assistance with organizational and cultural transformation.
- Drivers for organizational integration.
- Gain advantages over competitors that do not optimize resource management.
- Evidence that your business is a socially conscious one.
- Reducing Carbon emissions

3.3.4. PDCA (Plan-do-check-Act) model in ISO 50001:2018

Plan: In order to prioritize different prospects and chances for improving energy performance, the organization must identify the key and vital energy applications during the planning stage.

Do: The stated energy objectives and operational controls have been defined and implemented during the Do phase. The top management provides support functions, and different roles and duties of EnMS are established.

Check: This step helps the company to assess whether desired results were obtained. Internal audit can enable a company to confirm that the EnMS is working properly and achieving the desired outcomes.

Act: The management evaluates the performance gap or need for more improvement based on documented internal audits, management review inputs, compliance evaluation, etc.

3.3.5. Planing of EnMS:

3.3.5.1. Energy review

- Based on the measurement and other data, calculate the use and consumption of energy.
- Identify the many types of energy used today.
- Analyze your energy consumption patterns, both past and present.
- Based on the analysis, identify Significant Energy Use (SEUs).

3.3.5.2. Energy baseline for collection of energy data

Data is essential for evaluating and continuously enhancing energy performance. It is vital to plan for the type of data to be gathered, where the data needs to be collected, how the data will be collected, and how frequently the data must be obtained. It facilitates confirming the data availability required to have the energy review processes and the processes related to the measurement, monitoring, analysis, and evaluation. Data can come from simple numerical counting systems or from more complex monitoring and measurement systems coupled with applications that can combine the data and even do analysis automatically.

3.3.5.3. Operational planning and control

Regarding its SEUs, the company must develop, implement, and regulate the EnMS processes. Establishing standards for the procedures that consider the efficient operation, management, and maintenance of equipment, machinery, energy systems, and processes that use a lot of energy because their absence can cause a big deviation from the expected energy performance.

3.3.6. Assessment criteria

The organization must evaluate its environmental performance and determine EMS efficiency level.

3.3.6.1. Evaluation methods

- Identify compliance (occurrence, procedures and evaluation)
- Take any safe and structured actions.
- Document compliance evaluation information

3.3.6.2. Measuring and monitoring

- measurement in relation to goals
- Development of mechanisms for ongoing improvement
- Monitoring of water discharges and emissions
- Energy consumption data
- Trends analysis
- Overall performance of the EMS

3.3.7. Requirement for energy audit

Internal audits of an EnMS can be performed by companies' employees, or by outsourced professionals appointed by the company and their work on its behalf. Auditor independence can be proven by showing that they are not responsible for the activity under audit. An energy audit or energy assessment is not the same as an internal audit of an EnMS.

4. EMS Networks and Functionality

The term Energy Management System in the context of this report references to a system that focuses on the energy consumption of a specific building primarily but can also encompass minor energy production and storage like solar panels and in-building batteries.

There is a high variety of these types of energy management systems in the current market. There is a limited selection of free and open-source options, but most of them are products sold by providers within the field. These aim to largely cover the same primary goals outlined in the ISO 50001 standard, briefly summed up to monitoring energy consumption to keep managers continuously aware of the energy performance of the building. Some providers go further and have extra features to add more value to potential customers with active building managing, machine learning predictive modeling and renovation proposals. This chapter seeks to map the different tools and methods used in general terms and how they can be beneficial to achieve energy targets.

A system like this is often included in larger scale building management systems that has a much larger scope than simply energy. Building management systems can include anything from space management, room booking and calendar functions and camera surveillance. Because the scope of this project is limited to specifically energy management, and these added building management functions have little to any effect on the energy consumption of a building, these will not be evaluated, even though they may in some cases be inseparable from the energy management functionalities.

4.1. IoT (Internet of Things)

Internet of Things or IoT is a system where computing devices, mechanical devices, etc, are interconnected to transfer data and communicate. An IoT system consists of smart devices such as controllers, and sensors which are web enabled embedded systems. These smart devices collect data and send it to a connected IoT gateway which later processes data to the cloud for analysis. Some of the common IoT system are Zigbee, BACnet, etc.

4.1.1. Zigbee

The control and monitoring of linked devices is made possible through the international wireless radio communication standard (IEEE 802.15.4) known as Zigbee. For home automation, it is a low-power, near-field solution.

4.1.2. BACnet

Building automation and control network (BACnet) employ the ASHRAE, ANSI, and ISO 16484-5 standards protocol to building automation system. Building automation and control systems, such as HVAC, lighting, access control, and fire detection systems, can communicate with one another thanks to BACnet, a technology that was developed specifically for this purpose. Regardless of the specific building service that each computerized building automation equipment delivers, the BACnet protocol offers channels for information sharing and data analysing [35].

4.2. Data Collection

One important requirement for an energy-efficient system is data collecting. When gathering the information, an energy management system need to be adaptable. An EMS should smoothly integrate and retrieve such data from spreadsheets, old metering systems, or building automation systems for analysis [36]. The use of historical data is crucial in energy management systems. It helps to see energy patterns, compare current energy use to earlier years, and identify prior issues.

4.3. Sensor implementation

Sensor implementation is one of the key functions of EMS. Nowadays, sensors play a key role in several electrical and mechanical equipment. They are converting the functionality of gadgets into intelligent systems. A sensor serves a distinct and well-defined role in every system. In order to provide higher-level and more precise information, sensor fusion algorithms integrate the data streams from each individual sensor [37].

4.4. Energy Prices Forecasting

Energy forecasting is a method for estimating future energy requirements to reach supply and demand equilibrium [38]. The goal of the price forecast is to provide a solid foundation for the price modeling used in long-term energy planning [39].

5. Consumer Software

Consumer software is a category of energy management system that is sold or available for free separate from any installation and management services. This type of EMS is mainly marketed towards home installation but can also be effective on small to medium-sized office buildings.

5.1. Home Assistant

It is an open-source tool for home automation. It allows the user to control remotely many devices in the house and monitor sensor readings. With Home Assistant, the home-owner can identify which lights are on and turn them off, they also can know the temperature in the room and establish set points for it. [17]

5.1.1. Installation

One of the main advantages of home assistance is its openness. The main control unit can be installed and configured in different devices, such as: Raspberry pi, Odroid, Asus Tinkerboard, Windows devices, macOS devices, and Linux devices. Once the Home Assistance OS is installed in the local device, it is possible to access its interface using a web browser.

5.1.2. How It works

Assuming that the Home Assistant OS was installed in a Raspberry pi, once this device is located at home and connected to the wireless network, the user can connect to it through a web interface in their laptop or smartphone. Once in the interface, the home assistant will locate other smart devices that are connected to the network. Home assistant has many integrations, including Google home, and Zigbee devices. If a device is not located automatically, it can be added manually.

Having the integrations in place, it will be possible to create automation scripts and/or control manually all the devices associated with Home Assistant. Finally, a very interesting feature is the energy distribution chart. With this chart it is possible to understand the house's energy mix, namely, what are the energy sources in the house and what their contribution to the house's demand is, as seen in Figure 5-1.

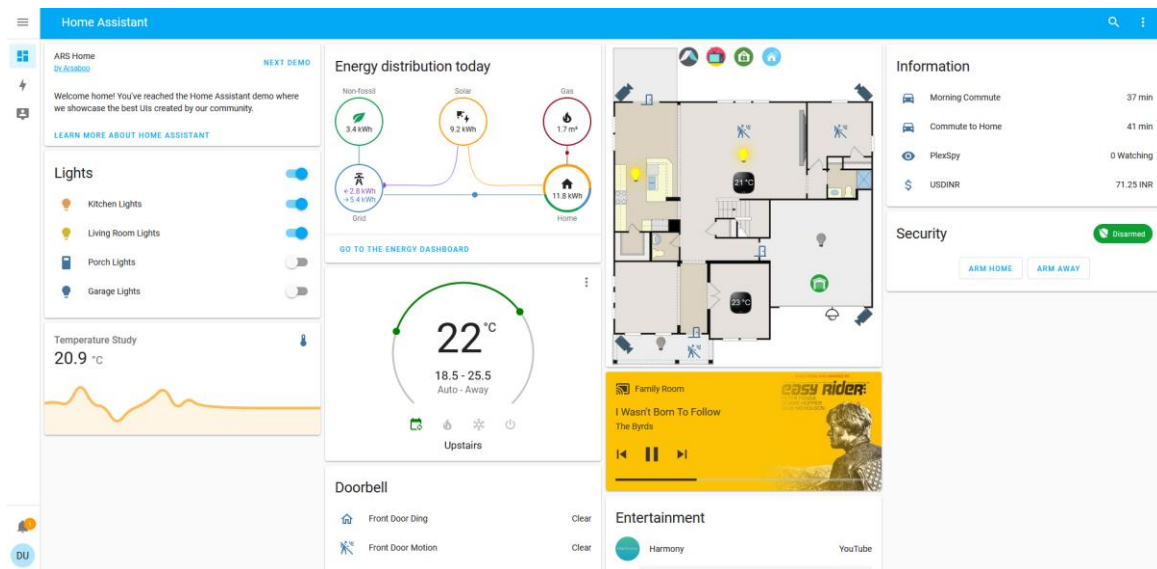


Figure 5-1 Home Assistant Demo Interface

It has a separate section for energy monitoring

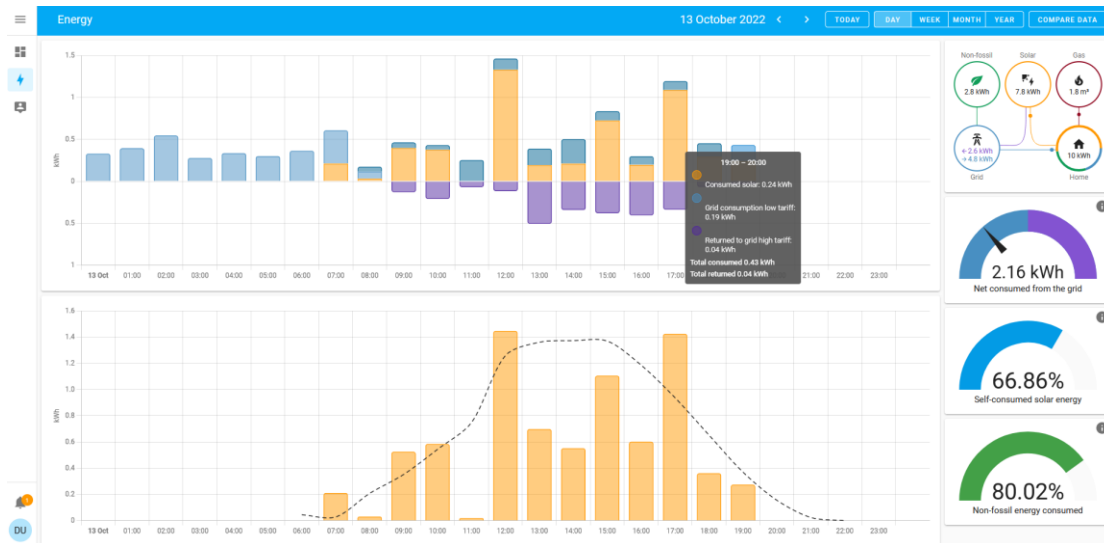


Figure 5-2 Home Assistant Demo Interface

The interface in Figure 5-2 shows the energy that was consumed in different periods: day, week, month, and year. It shows the energy mix, which helps the homeowner understand their consumption in high-price and low-price times, the solar energy produced, and how much of it was consumed or sold to the grid.

At the end it shows the totals in kWh and the cost it represents, as well as consumption per device. i.e: Boiler, washing machine, Electric car, air conditioning, lights, etc. As shown in Figure 5-3.

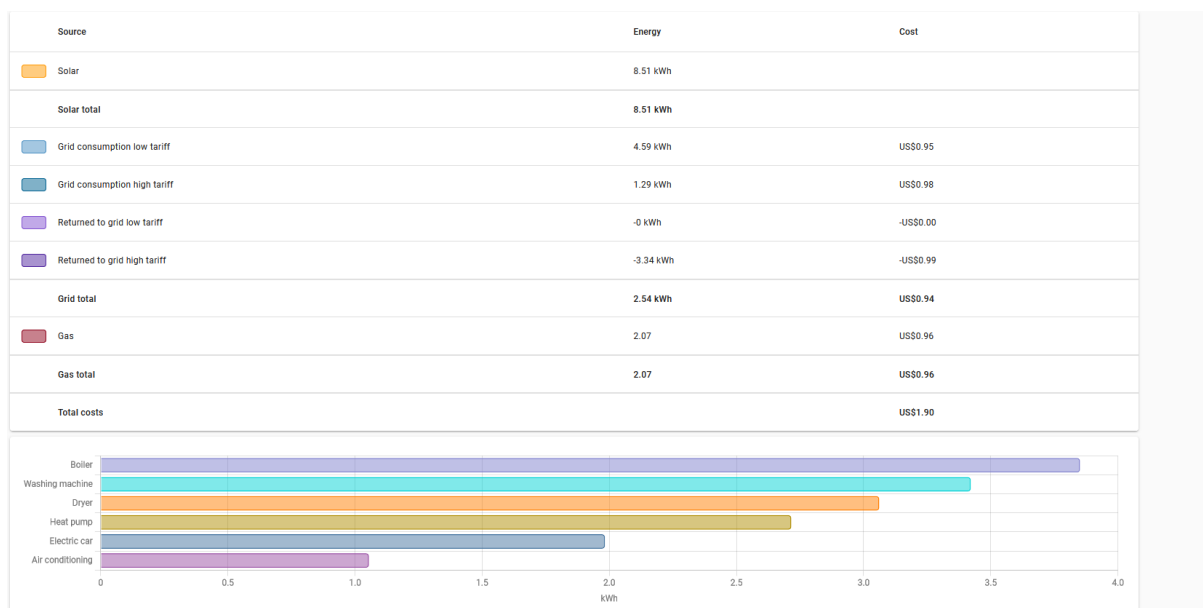


Figure 5-3 Home Assistant Demo Interface

A very important feature found in this system is the possibility to compare the current values for the current period (day, week, month, year) to the previous period, as shown in Figure 5-4. This gives insight into the tendency in energy consumption and how it is affected by possible changes in the house that occurred in between. For example, if between 2021 and 2022 there was a change to a more efficient washing machine, this will be shown in the year-to-year changes. With this, the homeowner will know which changes worked and what could be done to save more money.

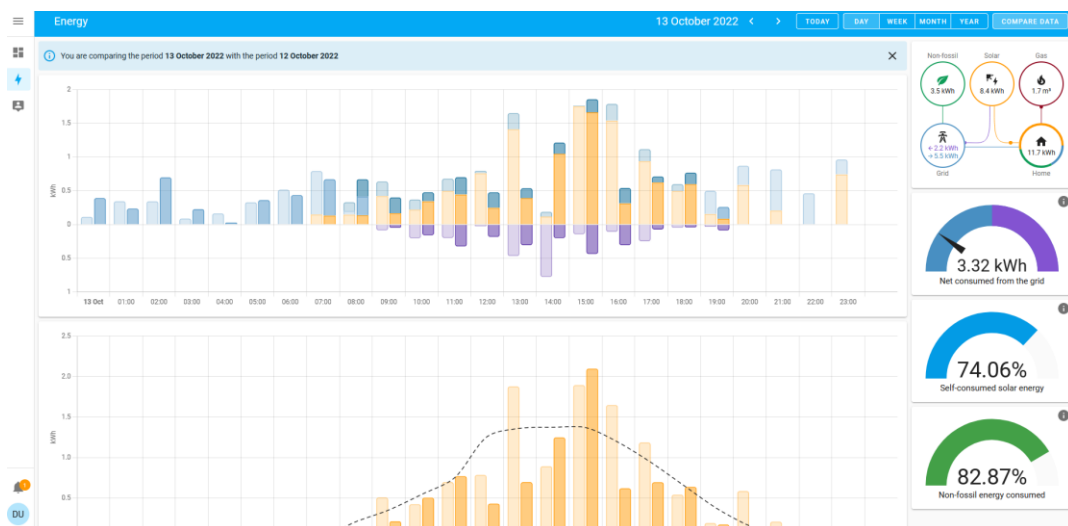


Figure 5-4 Comparison in the interface

5.2. Strømkontroll

Future Home is a product developed by a Norwegian company called Futurehome. It is distributed in Norway in collaboration with Enova. Every home that gets energy consumption

advise and installs an energy management system will get up to 15.000 NOK in support from Enova. [18][19][20][21]

Strøkontroll is focused in optimizing what they describe as the biggest energy expenses at a home: Water heating and floor heating. Their goal is to enhance energy savings by reducing the consumption when the energy prices are higher and by distributing the energy demand through several hours. The later could be described as a simple peak shaving method.

Physical Modules

The system requires hardware working in sync with the futurehome app, the following elements are portrayed in the scheme from Figure 5-5:

- Han Sensor: Reads the consumption data from the AMS meter, via the HAN port, then sends it to the smarthub for processing.
- Zigbee Thermostat: A smart thermostat that communicates via the Zigbee protocol. It will help control the temperature in the rooms.
- 16A Relé: It will control the water heater, it will help avoid heating the water when the energy is more expensive.
- Smarthub: It integrates the signals from the other smart devices in the house and sends it to the cloud, where the app can gather the information and display it to the user.

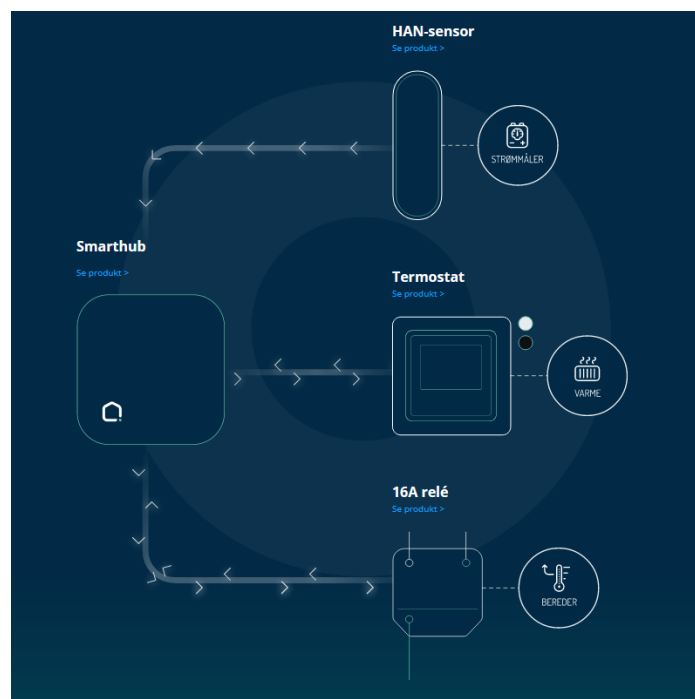


Figure 5-5 Physical Architecture, taken from

5.2.1. Mobile Application

Future home has an iOS and Android application that can be downloaded from any of those application stores, here is a walkthrough through the main functionalities:

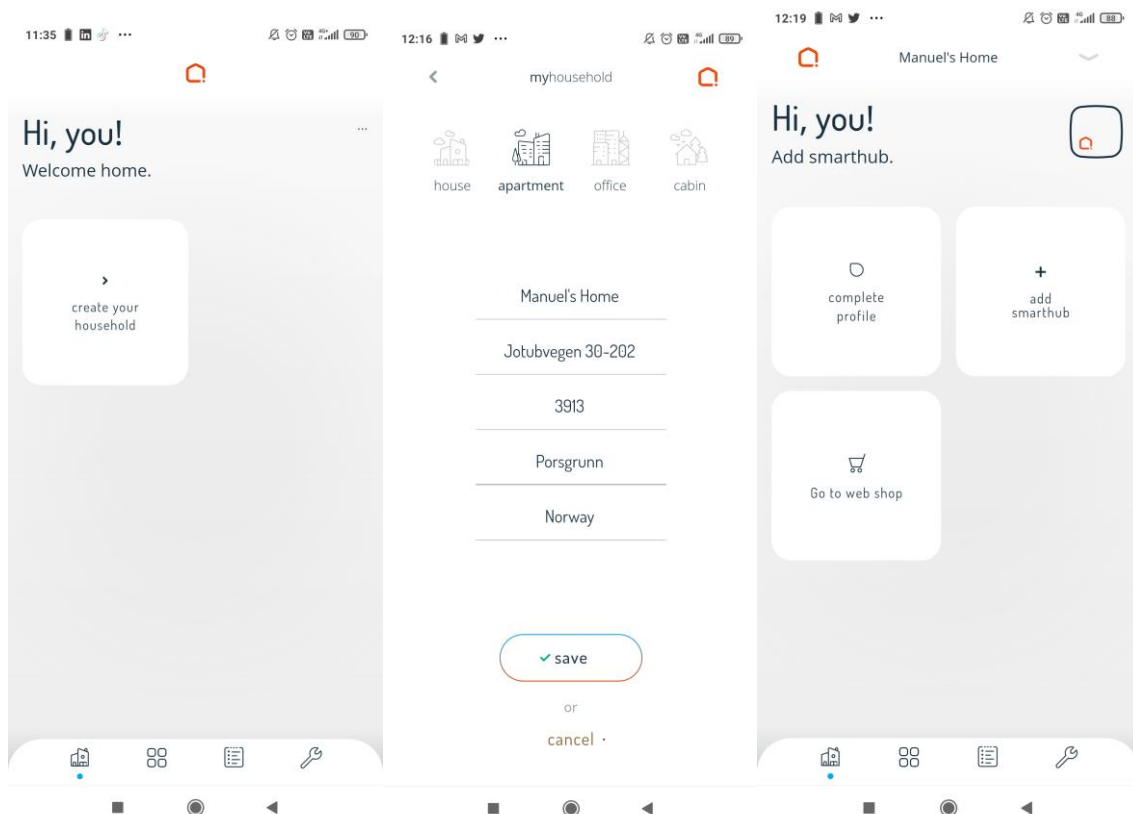


Figure 5-6 Futurehome interface – Screenshots from the application

In the android version, after install, the app prompt the user to create a home, with address and post code. Then it will request to add a smarthub, which would need to be bought for further testing, the screenshots are shown in Figure 5-6.

As there is no smarthub available, it is not possible to further test the application. However, based on some of the promo images and videos, it is possible to see that the application helps the user see their energy consumption in periods of time and control several smart devices inside the house. In Figure 5-7 it is possible to see an example of a functional user interface of Strømkontroll.

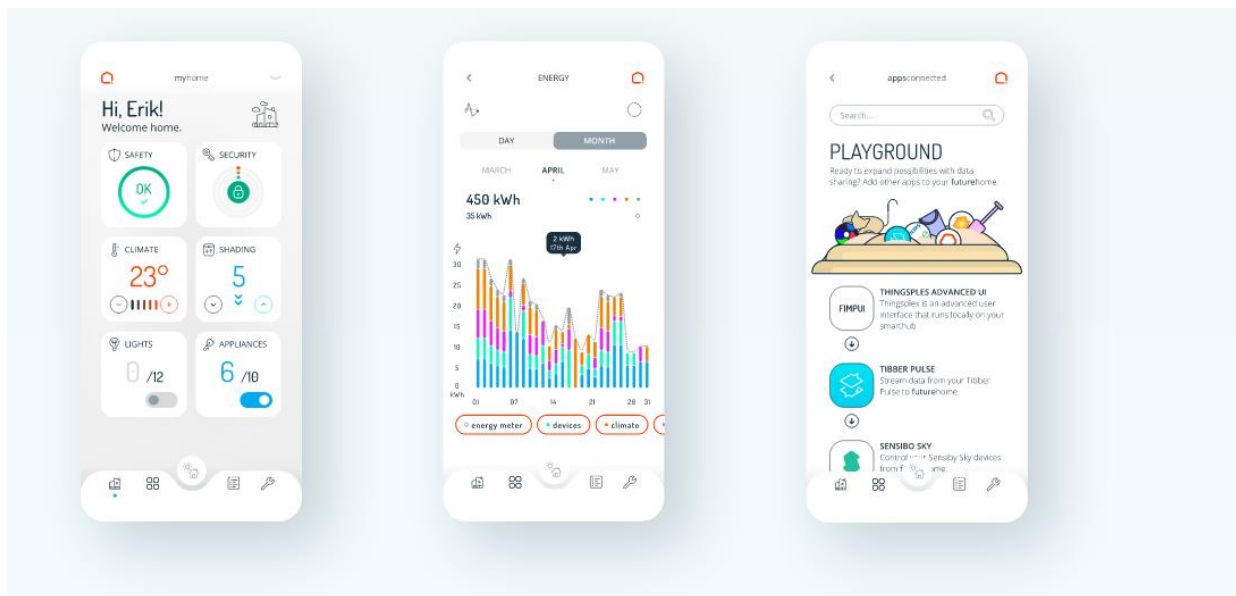


Figure 5-7 Future Home Interface

5.3. openMaint

OpenMaint is a software for property and facility management and related maintenance activity. It implements support functions for managing aspects of energy consumption. Its Energy Management module has the capability to analyze buildings energy information to highlight the energy consumption and meter readings.[22]

The functionality of the Energy Management module: It can read and collect data from sensors or meters or from any IOT solutions and track energy consumption. It can monitor and read the data from the sensors and update the database to check the energy consumption of the buildings. It can also trigger the workflow according to the data it reads. For example, if openMaint is connected to two sensors that collect temperature data from outside and inside of the building. If the inside of the building starts to get cold and there is a heater running inside, openMaint will trigger its workflow and alarm the maintenance team to check the condition.

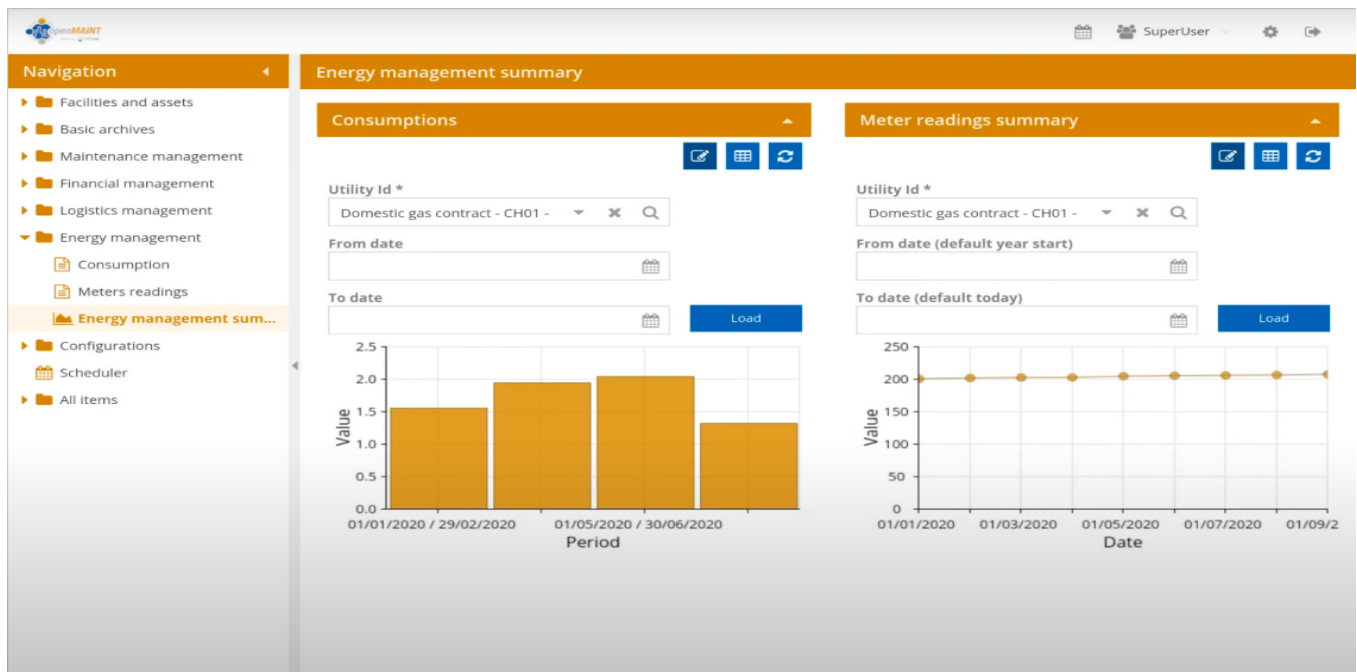


Figure 5-8 openMAINT Energy Manager dashboard

To provide a quick and convenient monitoring tool for energy management, openMAINT includes an energy management dashboard. A quick analysis of the relevant data may be done using several charts, including:

Consumptions: a bar graph displaying the amounts of consumption over the specified utility and time period for the relevant periods.

Meter reading summary: A line chart showing the readings trend for any utility (electricity, gas, water, etc.) for the selected contract and time period.

5.4. BEE2ENERGY

Developed by Compta Emerging Business (CEB solutions), BEE2Energy helps industries and institutions to achieve efficient use of energy while limiting the negative effects on energy consumption and lowering operating expenses. It is a cloud based IoT solution.

Overview: It provides all the necessary modules to help an organization easily analyze all the energy related aspects such as:

- Energy tariff plan management and simulation.
- Data collection.
- Controlling and programming associated devices.
- Forecasting and predictions.
- Alarms and reporting.

BEE2Energy has the capability to communicate with a wide range of equipments and devices such as: IoT gateways, meters, energy quality analyzers, actuators (smart or analog), environmental sensors (temperature, humidity, water, air quality).

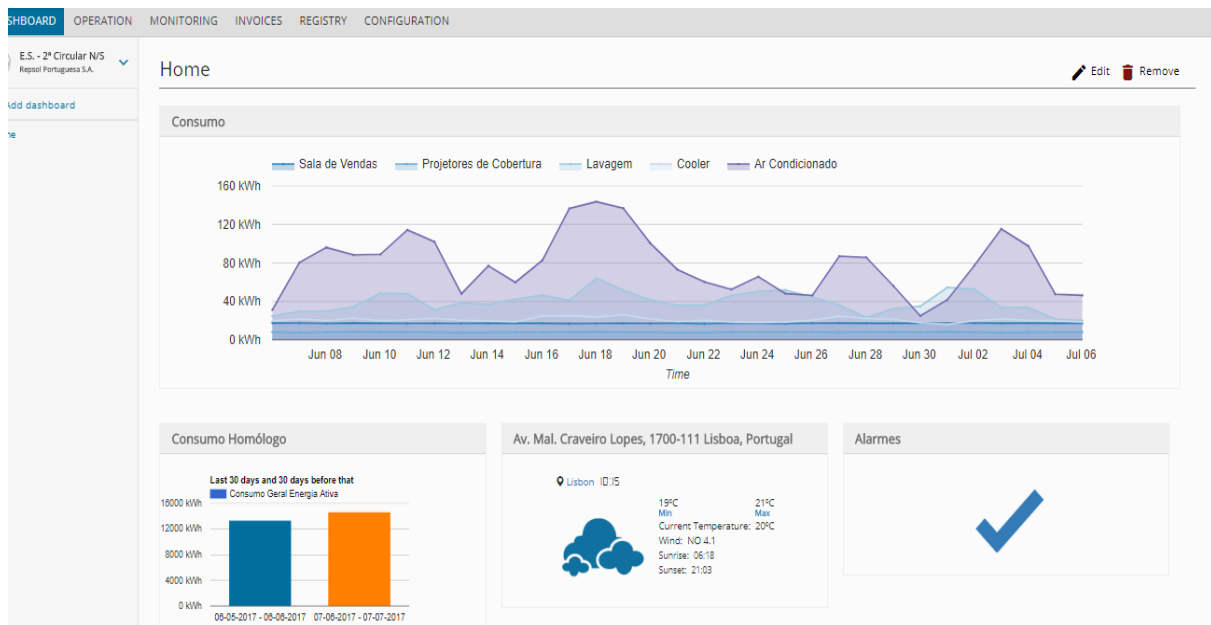


Figure 5-9 BBE2Energy Dashboard

BEE2ENERGY offers dashboard widgets and analytical tools that are simple to customize using an easy-to-use interface. It gives a quick and clear perspective of expenses, efficiency, alerts, and savings.

Real-time monitoring of usage, temperatures, humidity, and other indications is available with BEE2Energy. On the operation module it is possible to set up rules and programs with set-points to automate operations, and automatically turn on/off equipments. It is also possible to implement different criteria depending on sensor inputs for each location, system, department, level, or building. Additionally, it sends an email or SMS notification when the operation is executed.

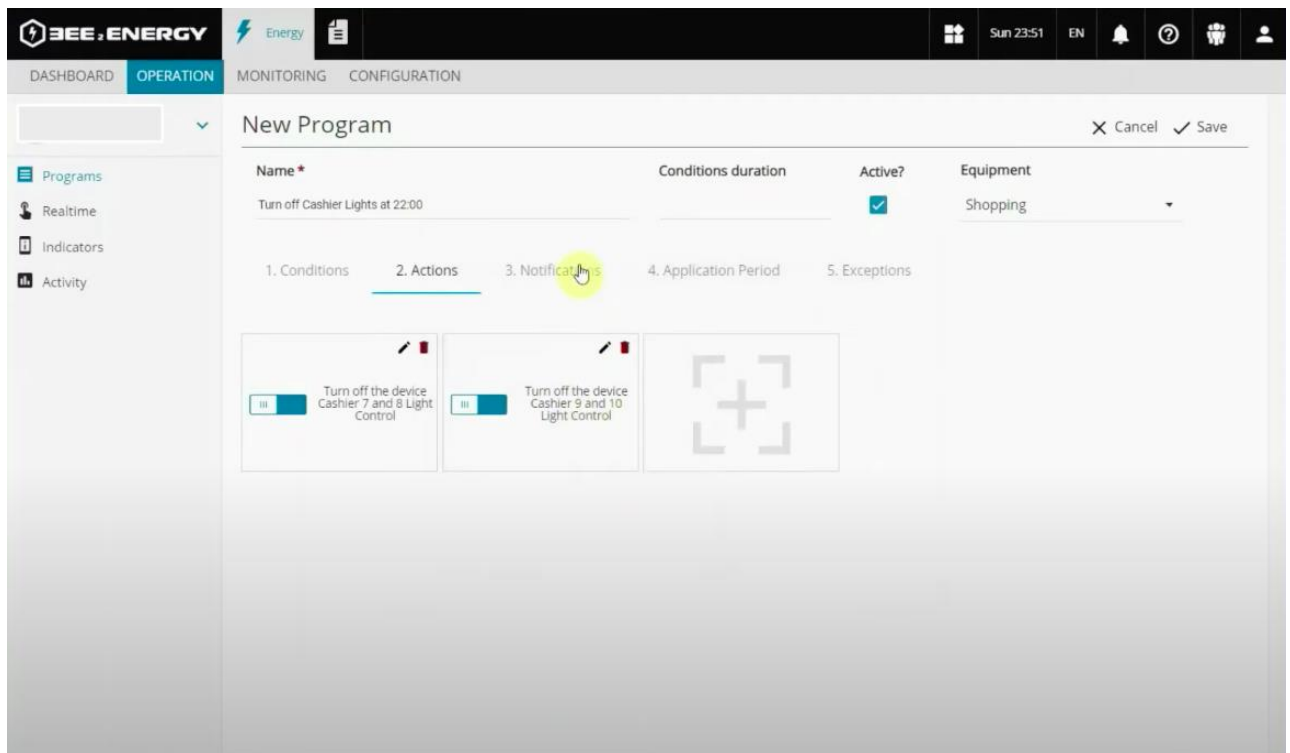


Figure 5-10 Advanced programming and operation module

5.5. Homey

This platform offers similar functionalities to those of Home Assistant and Strømkontroll. With this application, the user can control several smart devices at their home. It integrates a smart hub device that connects to other smart devices inside the house through wireless protocols, such as: Z-Wave, Zigbee, Wifi, Bluetooth, 433 Hz and others. [23][24]

Figure 5-11 shows an example of the user interface of the application.

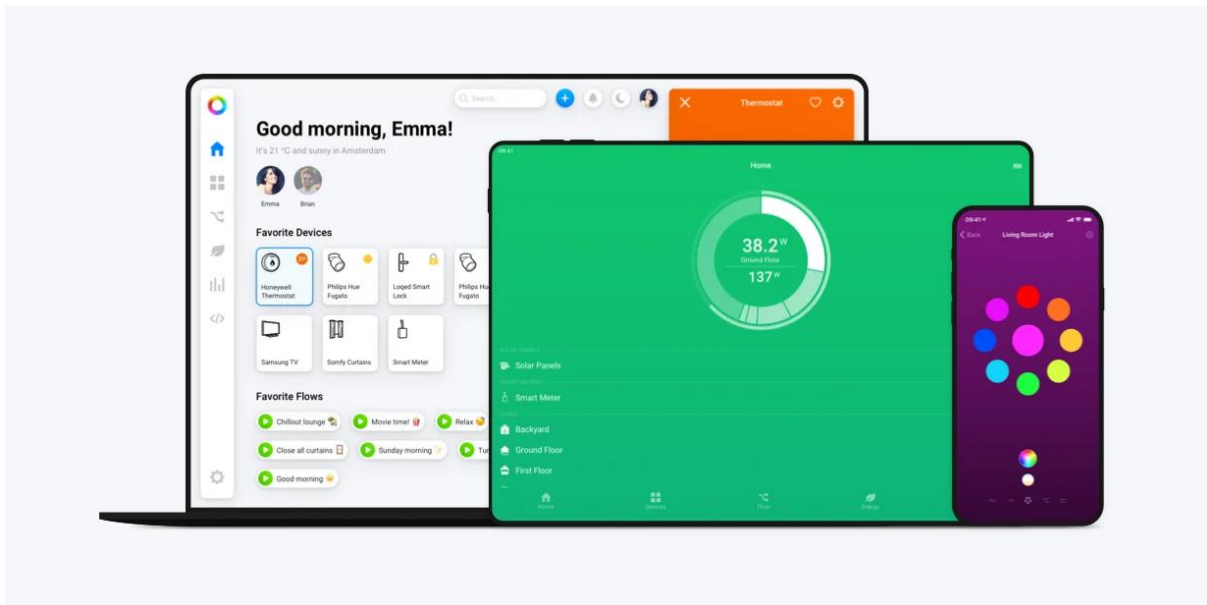


Figure 5-11 Homey Interface

Having connected to the devices, it is very easy for the user to control them through a functionality that Homey calls “Flows”. The user can download many openly available flows developed by other users and control all the devices at their home through the homey app. Some of the devices and applications that can be integrated with Homey are displayed in Figure 5-12.

This will let the owner of the house or building define when the water boiler should be started, when the lights of the house should be on and off, when to open and close the garage doors, and when to start the washing machine. The user will also be able to know when the house is empty and define rules for that scenario. Basically, any device using the protocols can be a source of information and used to define a control rule.



Figure 5-12 Homey Interface

The whole purpose of Homey is automation and while reading data from the devices and from energy prices sources such as Nordpool, it is possible to design many flows to optimize energy consumption. Homey has an open API, which facilitates the writing and importing of flows.

Homey Insights

Homey has a web interface that can be used to monitor the sensors readings a the house. It acts as the control panel from where the user can define temperature and consumption set points, also where it can start and stop the devices and get an overview of the control flows that their system has installed. Since is is needed to have a Homey pro device and sensors connected to it, the main dashboard in Figure 5-13 does not show much detail.

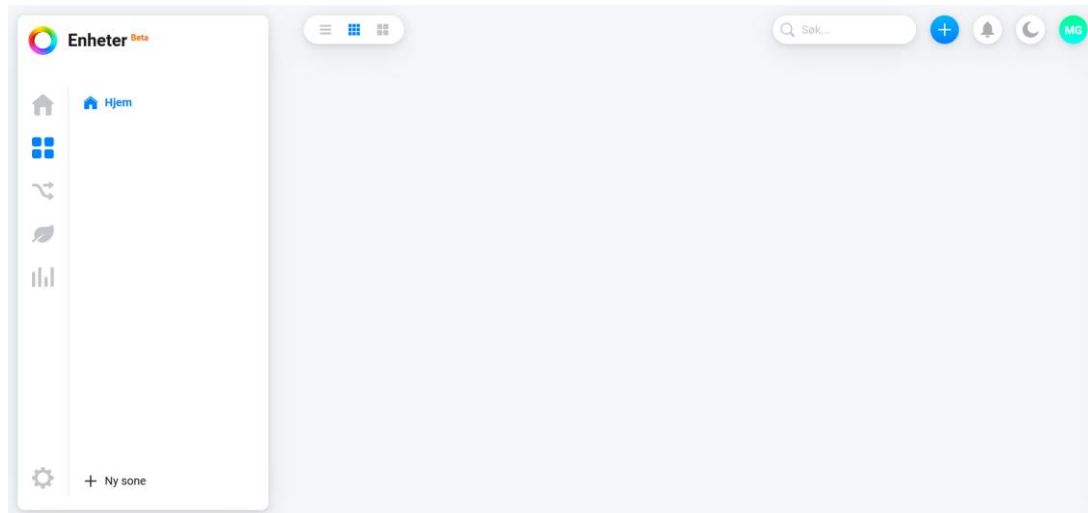


Figure 5-13 Homey Interface

5.6. RETScreen

About: In order to assess and ascertain the actual and ongoing energy performance of buildings, factories, and power plants around the world, as well as to manage portfolios of facilities, professionals and decision-makers can use the holistic Clean Energy Management Software platform known as RETScreen Expert. Over 750,000 users across every continent utilize the program, which was created by the Canadian government in association with eminent international partners. More than 1,400 institutions and colleges utilize RETScreen for research and teaching, and it is regularly mentioned in academic publications. [25][26]

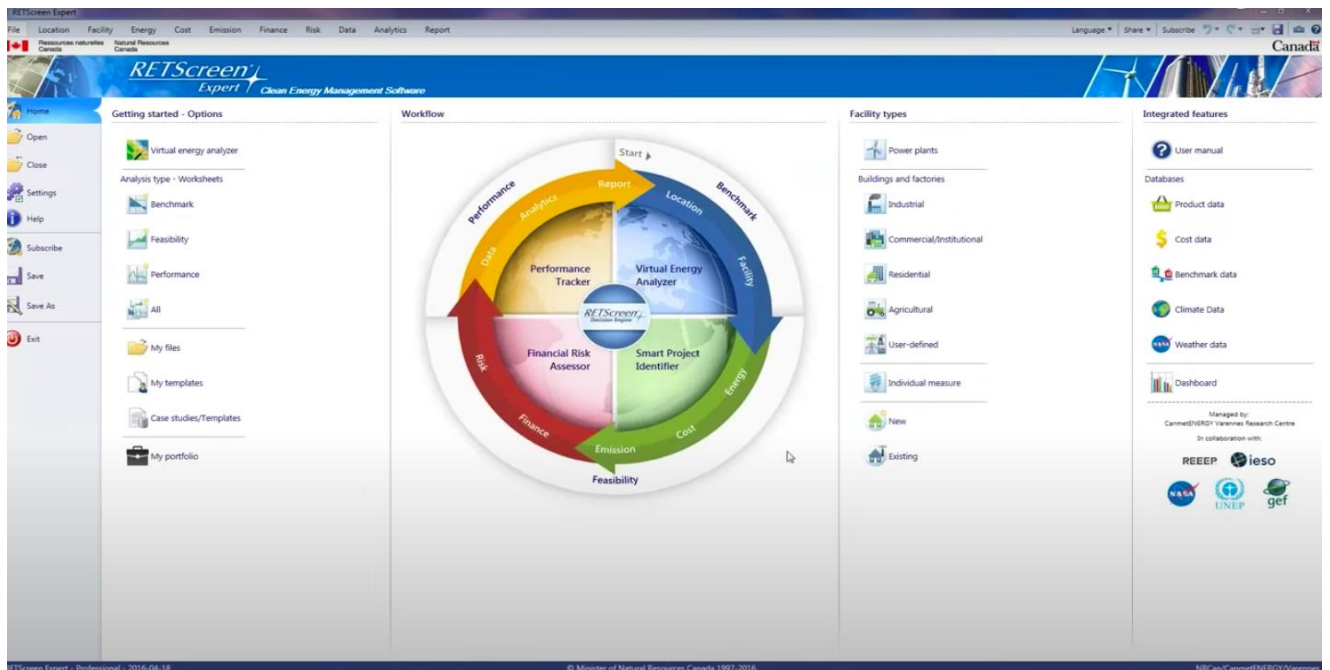


Figure 5-14 RETScreen main interface

The Benchmark Analysis function aids designers, facility operators, managers, and senior decision-makers to swiftly assess a facility's energy performance with the use of energy benchmarking. Helps estimate energy output, consumption, and potential for improvement.

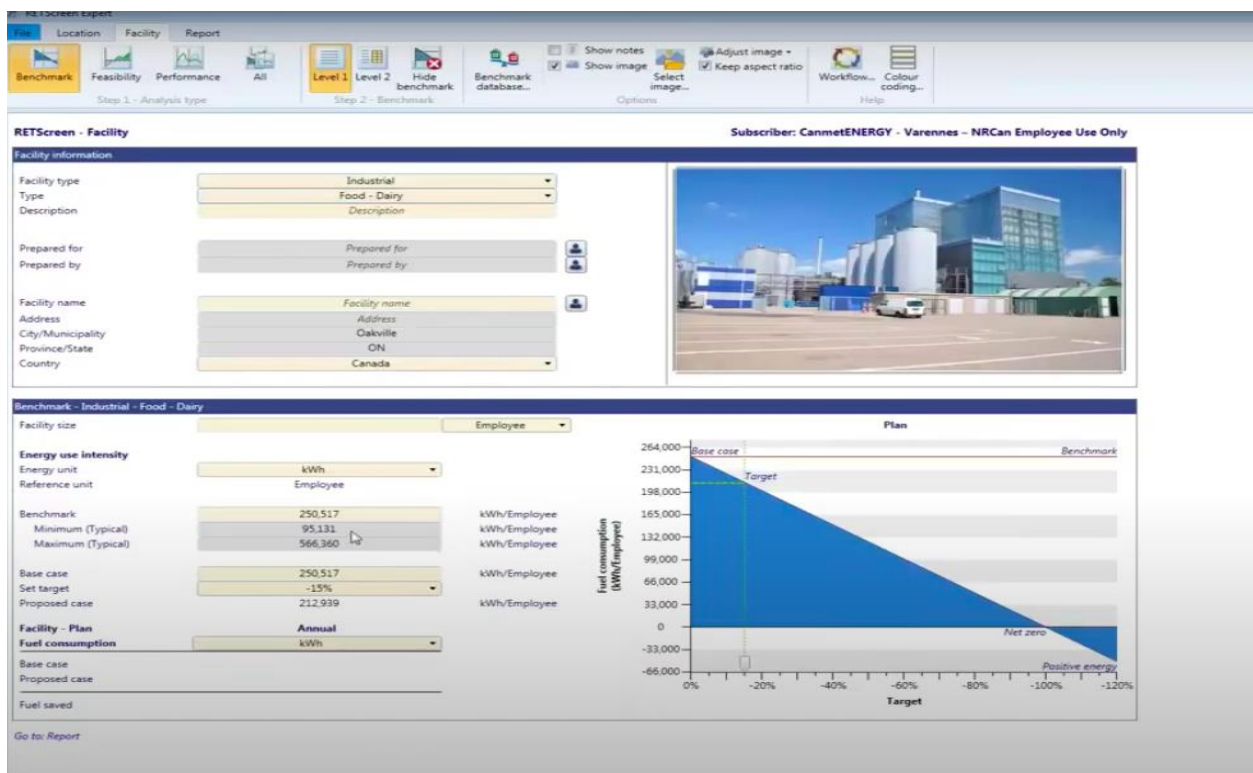


Figure 5-15 Benchmark analysis interface

The Virtual Energy Analyser feature enables quick and precise estimation of the energy production and savings potential for any place in the globe without the need for a site visit.

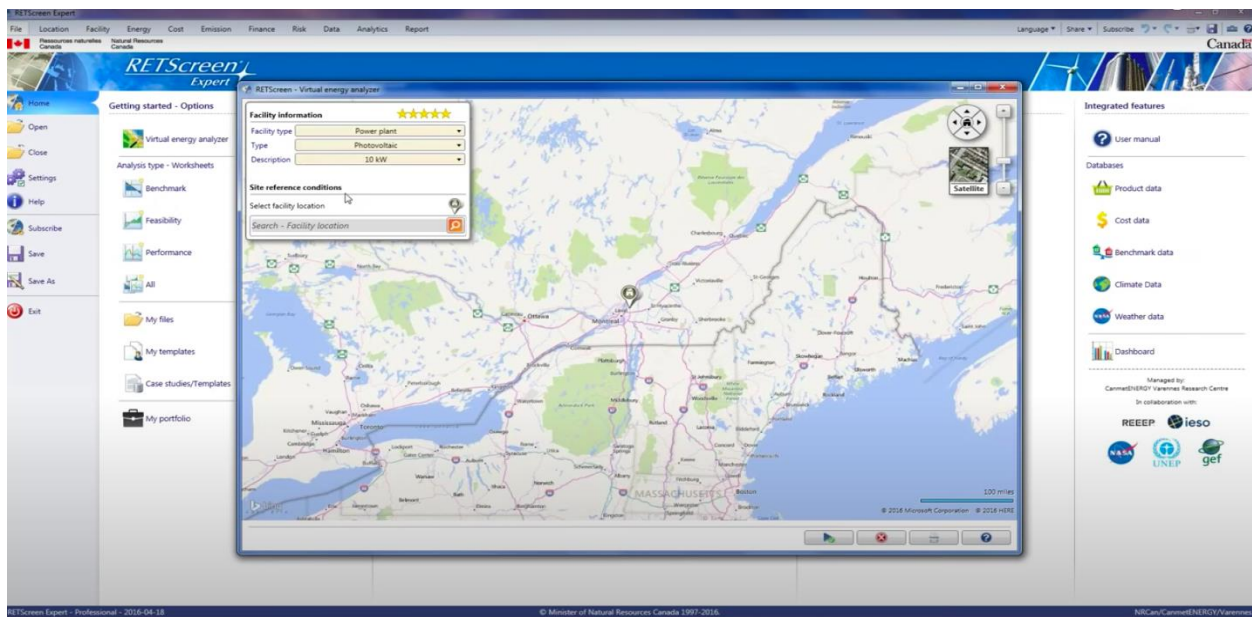


Figure 5-16 Virtual energy analyser interface

Feasibility Analysis: This function helps to do more detailed analysis on estimating energy production, energy savings and cost. Greenhouse gas emission analysis can be done by this module.

Performance Analysis: This function thoroughly evaluates and confirms the effectiveness of initiatives that have been executed, compares analyzed data with real data and assists in identifying potential for the facility to make additional energy improvements.

5.7. Energy Lens

Energy Lens is a free software and can be used for graphing and analyzing energy usage. It is used by energy managers and energy consultants to monitor and manage energy use. Energy Lens works as an Excel plugin and converts intricate, raw data on energy use into beneficial graphs and statistics. These graphs help to identify when and where the energy is being wasted and how much progress has been made in order to reduce energy waste.[27][28]

Software Interface and Functionality: In order to start analyzing, it is required to load energy interval data. As it is mentioned Energy Lens works as an Excel plugin, the energy interval data should be provided in Excel or CSV extensions. The interval data comes in two different formats either Row-per-day format or the Single-columns format.

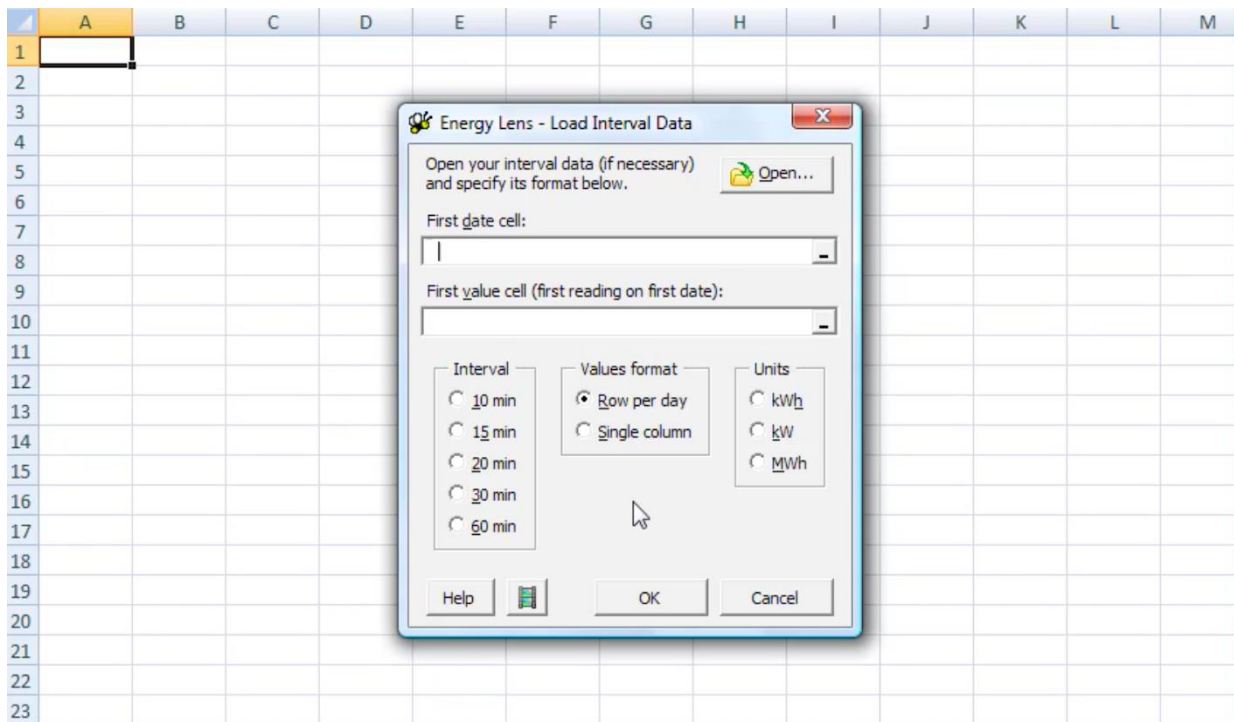


Figure 5-17 Energy Lens load interval data interface

Once the interval data has been loaded, a new interface will appear to start the analysis. The interface consists of different analysis of data such as daily, monthly, calendar, summary etc. And each of these functions works quite similarly.

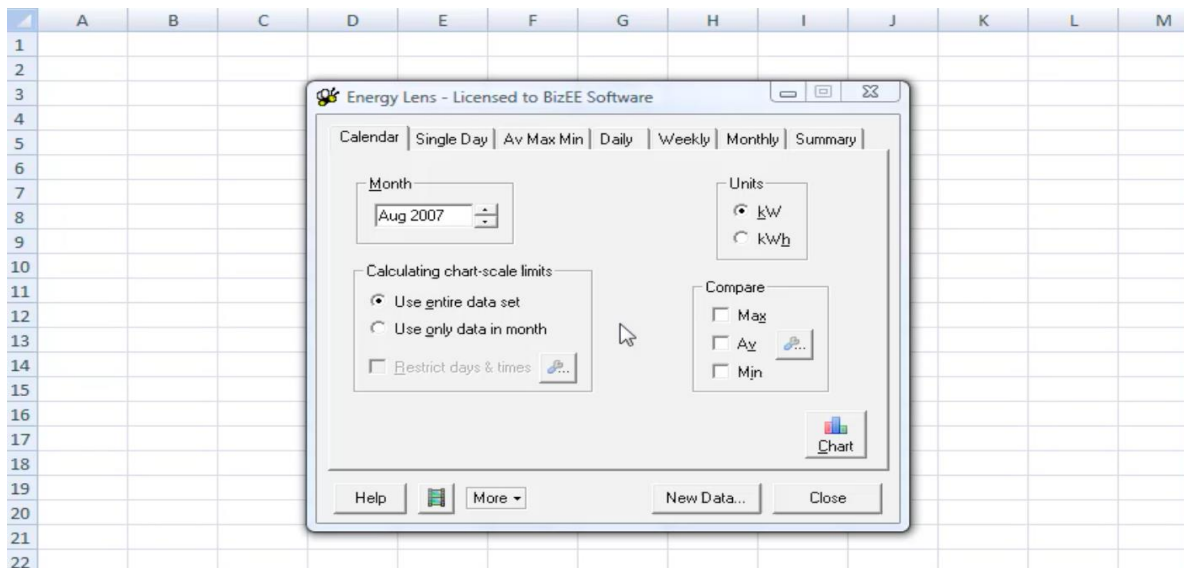


Figure 5-18 Energy Lens analyzer interface

Figure below shows how calendar feature helps us identify energy consumption and energy waste throughout the month. It can create energy chart which allows users to generate patterns of energy consumption. Each pattern represents single day, starting from 00.00am to 24.00pm. It gives rough idea about how much energy is being used at different times of the day. This plots are also helpful identifying energy wastes at different times of the day.

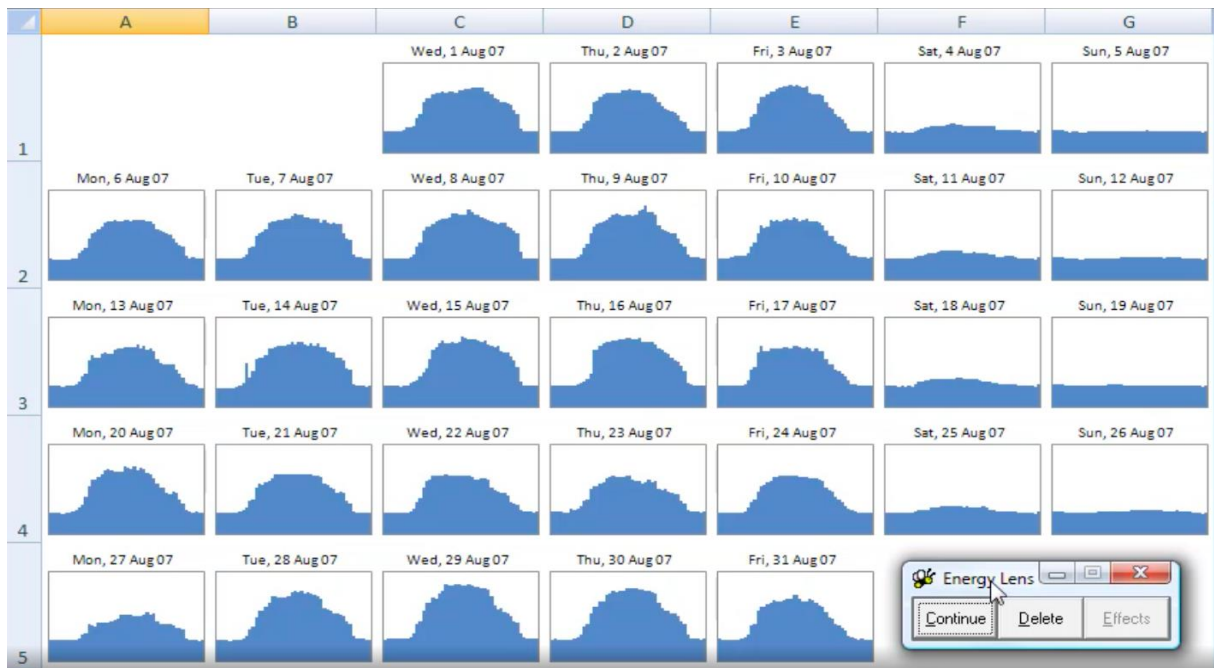


Figure 5-19 Energy Lens calendar feature interface

6.EMS Providers

Energy management providers are companies that sell, deliver, install and/or manage energy management systems for the customers.

According to the survey, which can be seen in Appendix A, the most attractive benefit of energy management would be reduced energy costs if one company was willing to apply it. Customers also prefer a system with a user-friendly design and visual information. The company was ready to provide information about ventilation and heating installation, as well as hot water supply. For the company, efficiency and energy optimization were essential. According to the answers to this survey, the most significant challenges to developing passive buildings are financial concerns and a lack of specific examples for companies.

As a result of the factors mentioned above, companies will be encouraged to invest in an energy management system.

6.1. Energihuset

Energihuset is a company located in Drammen, Norway. While they do not offer an energy management system or other software or hardware solutions, they do offer consultant services designed to cover a lot of the same problems energy management systems is designed to do. Energihuset's main selling points is to map a building's energy needs, consumptions and possible improvements, adaptations and upgrades for any specific building like most energy management systems providers, but because they do not offer an energy management system, they achieve this with expertise knowledge based on a building's existing building plan and technical data, historic energy consumption data, and local geography and regional climate.

They deliver this service as a one-time analysis of the building, rather than the continuous running service other reviewed providers in this report do. But if their marketing material is to be believed it does raise questions about the viability of these systems if much of their functionality can be achieved with existing data and expertise knowledge. [29]

6.2. Ant Solutions

Ans Solutions describes itself as an Industry 4.0 provider. They specialize in software solutions for manufacturers. They sell different modules that interact as an integrated system. One of those modules is an EMS, which for their case, not only focuses in energy consumption, but utilities as a whole. Such as: Electricity, gas, water, steam, compressed air, special liquids, and cooling.[30]

As shown in their web page, their EMS has four main functionalities, which are:

- **Energy Performance:** Provides a high-level view of the utilities performance of the whole manufacturing plant, with the option to go do a drill-down analysis, allowing the Energy manager to view the performance of an specific manufacturing line or machine.
- **Power Guard:** The Energy Manager can set utilities consumption limits. When one of such limits is crossed, the system will generate actionable alarms.

- **Utilities vs Production:** Showcases the utilities consumption of every production line, this information can be compared with the production information.
- **Schedules Utilities Used:** Shows a forecasting of the utilities usage and the expected associated cost.
- **Excise Billing:** Shows the expected savings on excise as a result of the efficient use of energy.

6.3. Clevair

Clevair defines itself as a solution for control of the indoor climate in small buildings. According to the company, it does not require big and lengthy installations, it is supposed to work together with existing systems. Their technology helps the energy manager control the energy consumption while keeping in check the CO₂ emissions. This is achieved by automating the HVAC systems of the buildings.[31]

6.3.1. How They Work:

First, Clevair establishes an energy baseline. Clevair compares the energy consumption of the building per square meter against the Enova benchmark. This will tell them how much potential improvement in the energy consumption there could be.

Depending on the facilities that the customer already might have, Clevair will install the equipment required to automatize the HVAC systems. For example: If a building already has a Siemens control unit, Clevair will install a device compatible to control and read signals from it. Whereas if the site only has independent HVAC systems and no smart temperature sensors, they will install a control system and all the sensors that are required to obtain the signals that they require to do an energy mapping and to control the devices.

Data is gathered from CO₂, temperature, and occupancy sensors in the buildings, which then will be integrated with weather forecasting and energy prices information. Then, they will use machine learning and proprietary algorithms to optimize how the ventilation systems operate. Some smart CO₂ and temperature sensors will be installed where it is required.

The main goal is to adapt the control signals to the circumstances that are outside of the customer's control, like the weather and the energy prices. If a hot day is expected, then the heating will be reduced. If the energy prices are high during a time of the day, then it is better to heat the building when they are lower.

Their algorithm also takes into consideration the occupancy in the building. During winter, during office hours, the energy required to heat the interior of the building is less, but at the same time, the efficiency of the equipment is reduced because of a higher concentration of CO₂. All these considerations are put together in the control algorithm, where the optimization variable is the energy required to achieve the different setpoints in CO₂ and Temperature.

The Clevair device will be connected through the serial port of a PLC system that controls the Air Handling Unit. With the optimized parameters from the platform, the device will control the HVAC system.

Then, the web platform will allow the Energy manager to control the setpoints remotely, track energy savings and CO2 reductions, and to receive alerts. At the end, according to Clevair case studies' it is possible to see a decrease in energy consumption between 30 and 40 percent.

6.4. FEMS - FENECON energy management system

Fenecom develops a product that is more focused in balancing the consumption with energy production and storage. For example: A house has a water heater and the usual electric devices, as well as an electric vehicle. But at the same time, It has a PV system and storage. FEMS (Fenecon's EMS) will help the owner control the sources for their home's energy consumption.[32]

6.4.1. Interface:

FEMS offers a web interface that can be accessed through a web browser. It displays the power in the solar cells, the power usage, the energy available in the batteries, and the energy that will have to be obtained from the grid. Figure 6-1 shows a screenshot of their demo. However, it does not show much useful information.

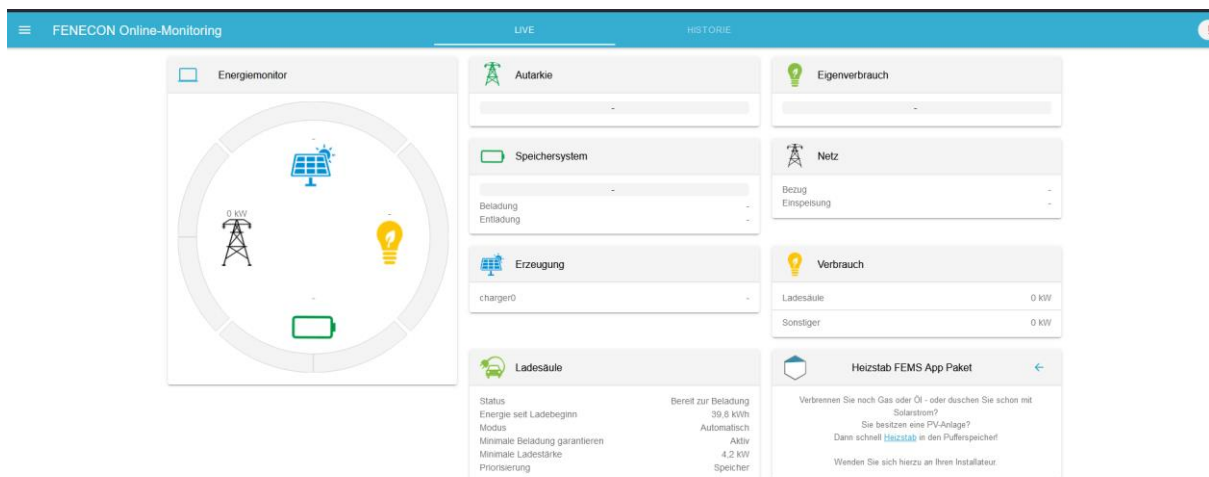


Figure 6-1 FENECON user interface

The interface also allows to obtain details about the power and demand sources and allows the energy manager to manually configure settings and levels.

6.4.2. Hardware:

To control all the systems, a device called the Fenecon box must be installed. It is described by the company as the heart of the customer's energy storage system. It is a relay board that has up to 8 channels to control devices and digital signals, an example picture is shown in Figure 6-2.



Figure 6-2 FENECON Hardware box

The box has domestic and industrial versions. If a customer has more than 8 devices to control, they can install more boards.

6.4.3. Control:

The system optimizes energy usage by dynamically controlling the generators and loads. It will use the charge in the battery during the night when it is expensive in the grid and charge it during the day.

It also has the possibility to do peak shaving, which is very interesting for the industries, where they have to pay additional fees for high demand peaks. If these demand peaks are avoided by automatically using battery energy, the fees can be drastically reduced.

FEMS will also optimize according to the energy tariffs. For this, the system implements prediction algorithms that estimate the energy consumption in the next 24 to 36 hours. If during this window, there is a time where the self-production capacity cannot cover the demand it will optimize the usage so that it takes energy from the grid when the price is cheaper. For this, it is important to have access to the prices via the interface of the energy provider.

6.5. Siemens – Sentron Digital

Sentron Digital is Siemens's energy management solution for buildings and small industries. Rather than a single application, they sell the solution as a holistic integration between smart electrical devices capable of communication through internet protocols, and visualization software that can be installed in multiple computer devices, as well as be used in the cloud. Siemens emphasizes their compliance with ISO50001, with the purpose of always having power consumption in view.[33][34]. Figure 6-3 portrays an example architecture for a Siemens implementation in retail.

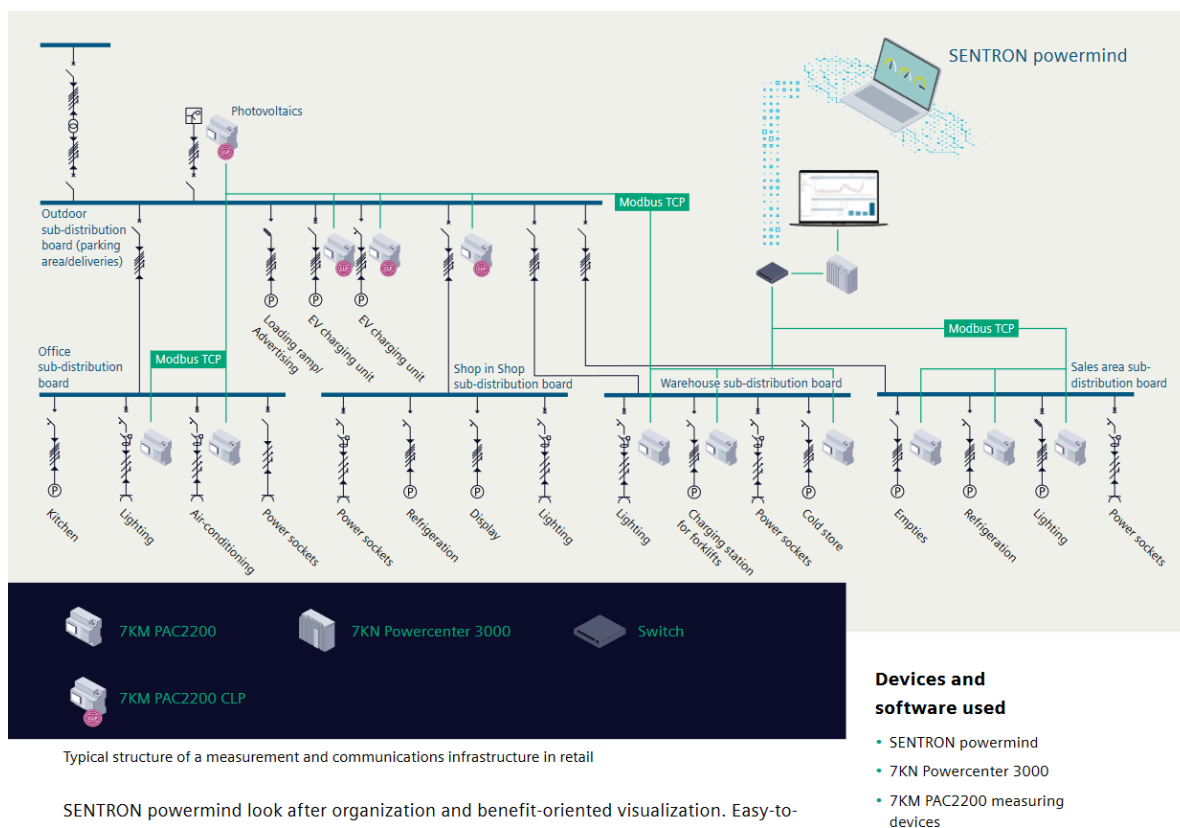


Figure 6-3 SENTRON architecture for retail

This brand designed their solution to give the energy manager insight in different levels, for each level of supervision there are different products:

6.5.1.7KN Powercenter 3000

This is a physical device that can be installed in the cabinet. It will gather the signals from up to 32 communication capable devices that are in the same location. The devices make part of the SENTROM line, these can either be: Energy meters, circuit breakers, circuit protection devices, virtual meters, and devices from third party vendors.

This technology allows the energy manager to make modifications during the commissioning stage or during operation, such modifications are the creation of zones and devices, as well as their deletion.

The data will be gathered and organized in a logical way by the 7KN Powercenter 3000 module and made available for visualization through the local network via a web interface or through a web application, this can be seen in Figure 6-4.



Figure 6-4 7KN Powercenter 3000 Web Interface

The energy manager can select to visualize power consumption in different time ranges, archive the data, create reports, and define alerts.

Data can be stored for up to 14 months.

The 7KN Powercenter 3000 used Modbus TCP to connect to the subordinate devices, Ethernet with Https for the local visualization, and MQTT for the cloud access.

Siemens Provides a free demo of the local visualization for this piece of equipment. It is possible to see the performance and the energy usage of different devices, that can be assigned to different locations in the user interface. The variables values can be seen in 15 minutes or in 10 seconds intervals. It also includes a log of generated messages, that can be sent by email automatically to the energy manager. The demo interface is shown in Figure 6-5.

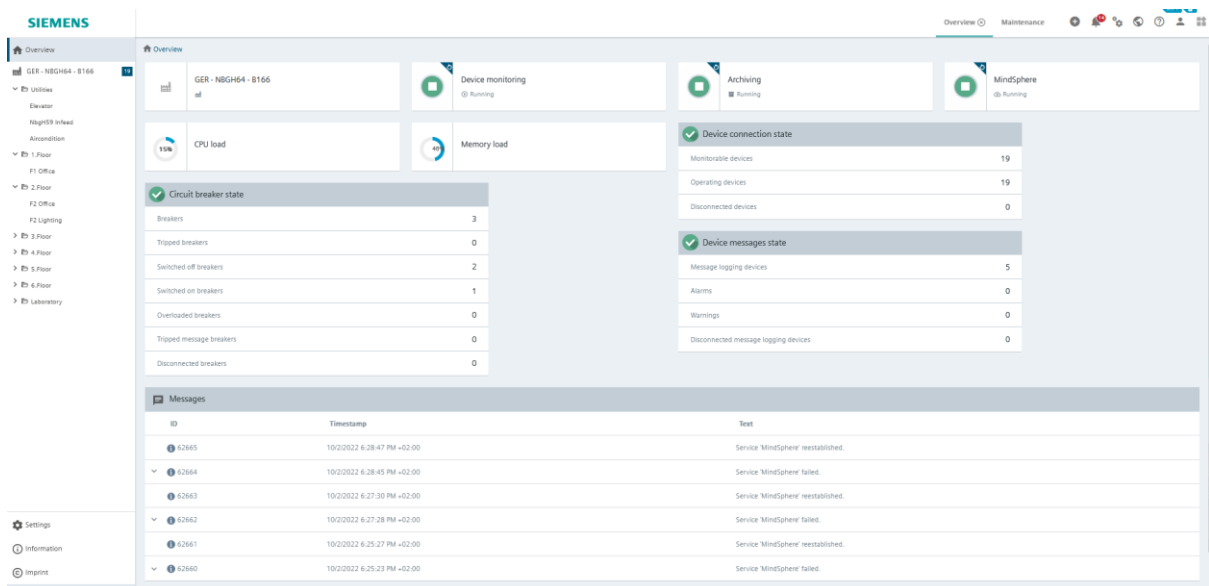


Figure 6-5 7KN Powercenter 3000 message board

6.5.2.SENTRON powermind

If an energy manager needs to control more than one building, a web application with high supervisory access is necessary, to aid this, Siemens developed SENTRON Powermind, which works in tandem with one or more 7KN Powercenter 3000 devices, which are installed in the customer's facilities.

SENTRON Powermind has two main purposes for the Energy Manager:

- Identify potential of energy savings by comparing consumption in different time periods.
- Identify and prevent the occurrence of peak loads by displaying the power consumption. As it has been discussed before, peak loads are an important situation to avoid for big energy consumers like public buildings and manufacturers.

SENTRON Powermind can be integrated smoothly with different 7KN Powercenter 3000 devices in a codeless process. With this network of 7KN Powercenter 3000 devices, connected to their respective smart measuring endpoints, and integrated in Powermind, the energy manages has a general perspective of the energy consumption in all the building under their supervision.

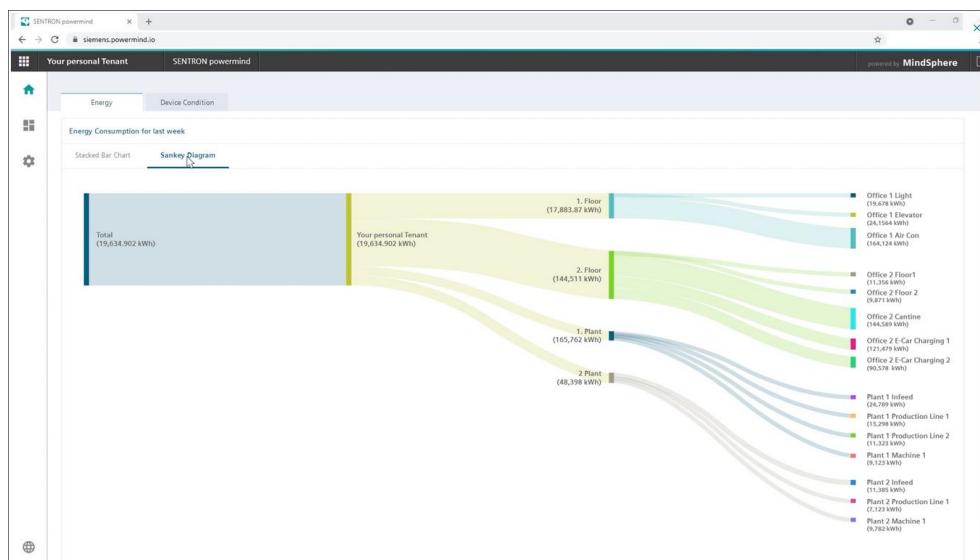


Figure 6-6 Sankey Diagram

The interface shows the consumptions of the metering devices and buildings in Sankey diagrams like the one in Figure 6-6. It helps the Energy Manager compare the consumptions of different time periods and visualize high consumption points through a heatmap.

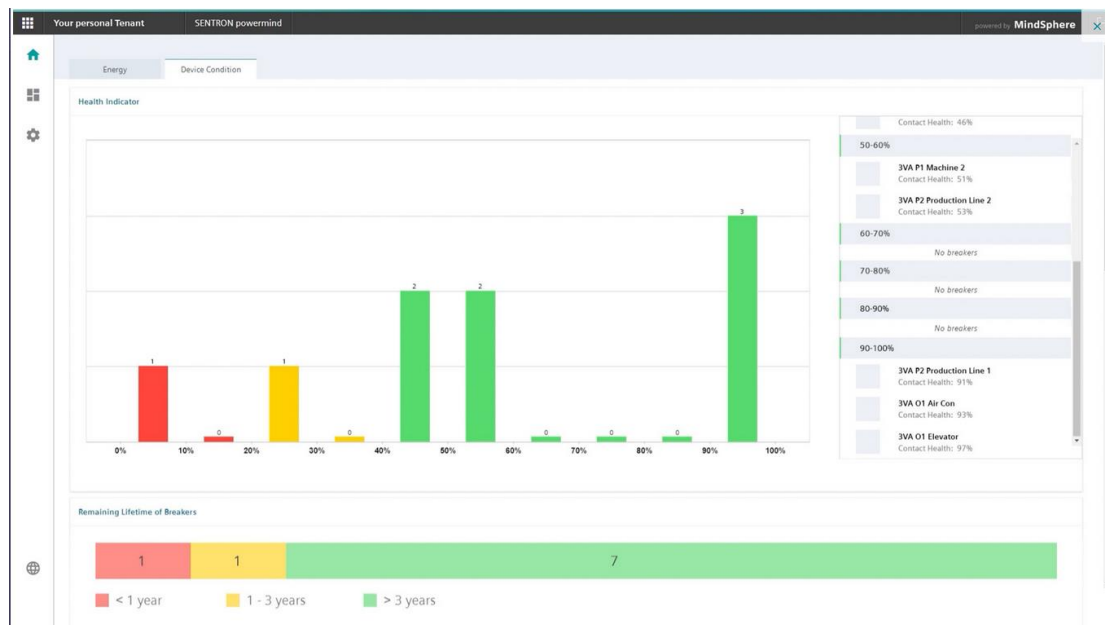


Figure 6-7 Asset Condition

It also displays the health status of specific devices along with their expected remaining lifetime. An example interface is shown in Figure 6-7. The system notifies the energy manager of abnormal and undesirable situations via push notifications and email.

6.5.3. Fleet Management

The web interface gives the most important information about multiple properties, with geolocation. It shows an energy efficiency icon, this is calculated comparing the current kWh consumption with historical data.

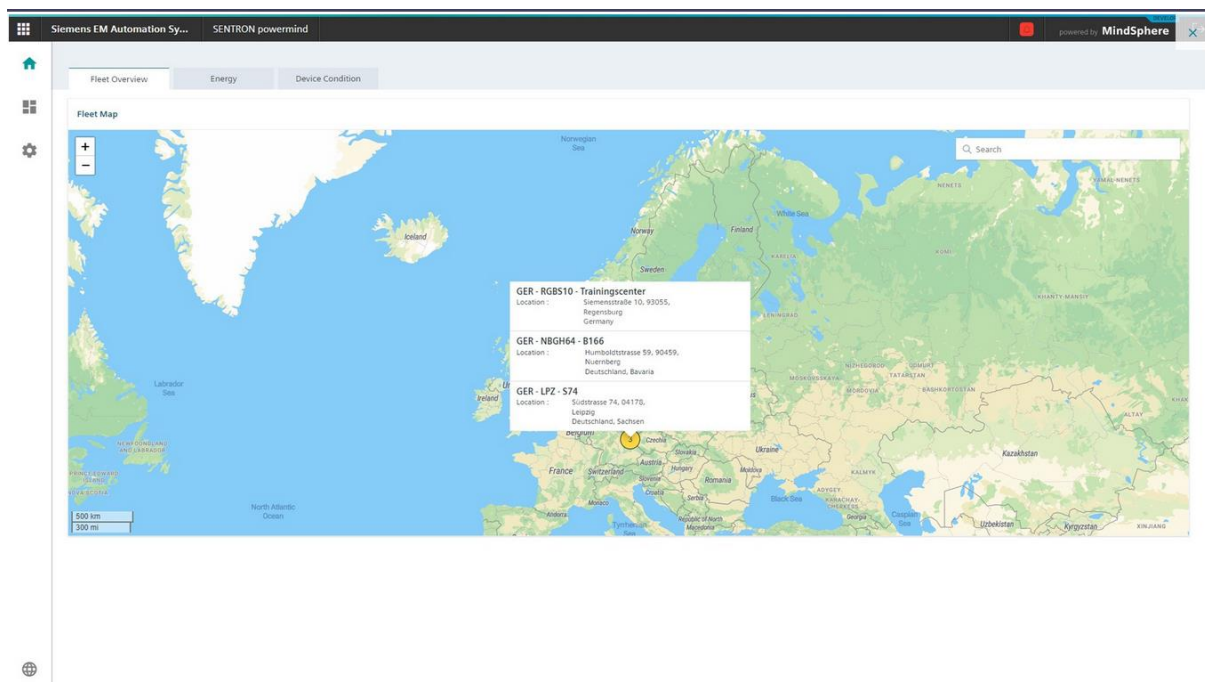


Figure 6-8 Fleet Management interface

It gives a quick overview of multiple properties.

6.6. Energy Manager AS – Real Case Studio in Sarpsborg Kommune

Energy Manager AS is a Norwegian firm which developed a large-scale energy management system. After requesting a meeting with them, it was possible to have a demonstration by Pål Berg, the CTO.

6.6.1. ISO 50001

Energy Manager AS develops its product around ISO50001. It is the core of the energy management. The application interacts with the organization in each of the stages of the PDCA model:

1. **Plan:** In this phase, the general aspects of the energy management system are gathered and assimilated: The team in charge reviews the regulations that apply and established limits about what setpoints in the operation of the equipment are allowed. E.g. Regulations could say that it is not allowed to drop the temperature below some value, which would put a limit in how much an HVAC energy consumption could be optimized.

In this stage the Energy review is done. During the energy review, the most significant energy losses are detected, for example: It could be detected that a building has several old lightbulbs and those should be replaced by LED lightbulbs, which consume less.

After having completed the energy review, the energy baseline is established, where the organization's goals in regards of energy usage improvement are laid out.

2. **Do:** This is the execution stage, where the changes that were planned are implemented, for example, the lightbulbs from the mentioned building are changed. This involved all the teams within an organization.

One aspect that is often overlooked is the importance of maintaining a balance between pursuing the energy targets and maintaining a viable business: A manufacturer should not have to stop their factory's production to reduce their energy bill. The viability of the operation must not be affected negatively.

However, if proper limits and considerations were established, there is no limit in what can be changed in an organization. Sometimes the projects will redesign complete systems, not just tune them.

Lastly, the procurement process is of significance in the success of the EMS. The requirements for the suppliers will be set up. For example, the organization could decide that it will only buy lightbulbs from an ISO50001 certified partner.

3. **Check:** Supervision, measurement and analysis. The energy manager gets an overview of the whole system and knows whether the plan is advancing as expected or not.

With the visualization and reporting tools, they will be able to detect deviations, set up actions, and see or raise alarms.

A good energy management system will show information in a context. For example, a machine consumption pattern can be affected by ambient temperature changes; perhaps the building where it was installed has new insulation, which causes the temperature to increase because the control algorithm of the HVAC was not reconfigured to account for this.

An aspect that sometimes is overlooked, is that often there will be conflictive objectives within the system: A ventilation system does not manage only temperature but CO2 levels, then, even if a room is within acceptable temperature values, if there is an increase of CO2, it will have to be activated, which will make it consume more energy to preserve the temperature levels.

4. **Act:** ISO requires that the upper management do an annual review, this way, it will be possible to determine how effective the investment was, take corrective action on non-conformities and tune the systems.

6.6.2. The EMS Software

Energy Manager AS provided us access to the implementation made for the Sarpsborg Commune. The commune uses it to monitor and control the energy consumption in all the public buildings and services under their responsibility. It is also used to monitor the facilities in Grålum and Skjeberg districts. It allows for the user to go from a very high-level perspective to a very low one. for example, it Is possible to see the energy consumption of the commune for a whole day, week, month, quarter or year, and to see the current ambient temperature inside a classroom in a specific school during that same period of time.

The main screen displayed in Figure 6-9 shows in a map the locations of the sites that are being monitored, along with a chart listing Norway's energy zones and the mean temperature at Sarpsborg. It also shows a summary of the active projects.

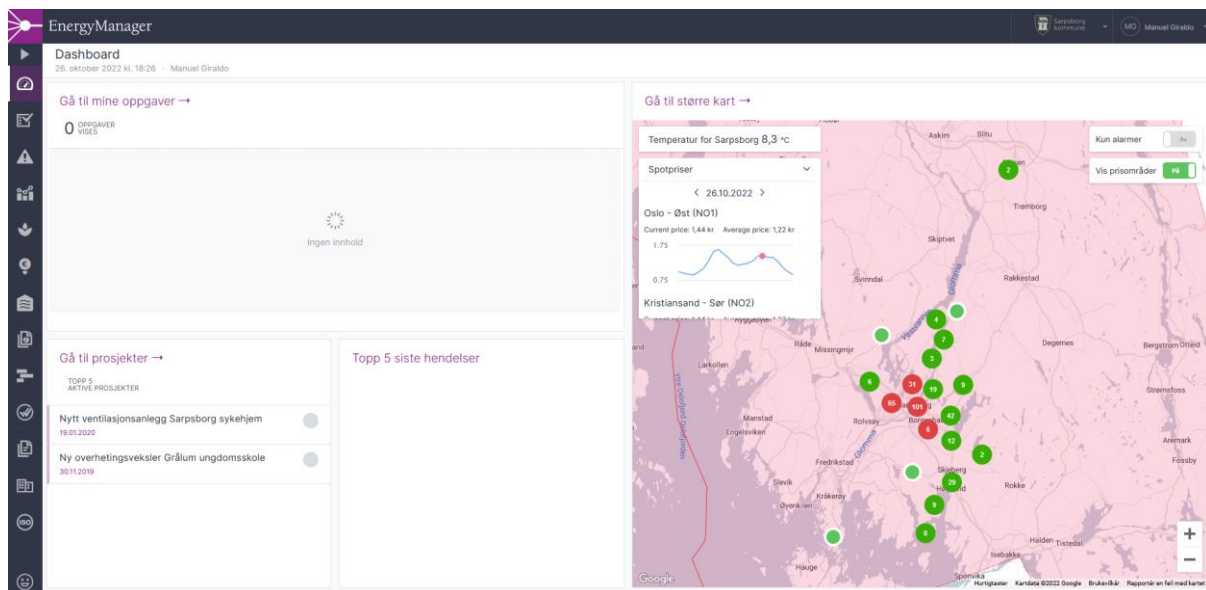


Figure 6-9 Main Dashboard of Energy Manager

As shown in Figure 6-10, The menu has several options: Tasks, alarms, Energy Follow-up (To view the energy usage at all levels), Environment, Costs, indoor climate (It is possible to view the inner temperature in different buildings), reports, Projects, Quality system, documents, building and equipment, and lastly, and ISO section, where the Energy manager can find everything related to the ISO5001 guidelines that applies for the organization.

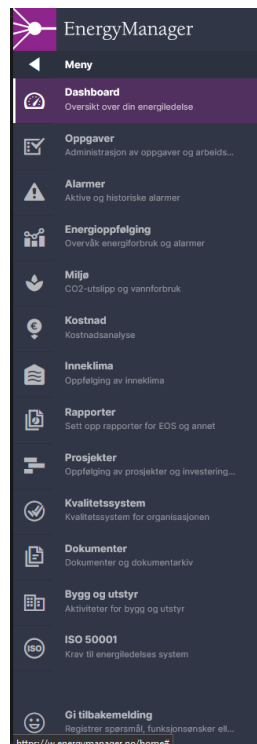


Figure 6-10 Energy Manager Menu

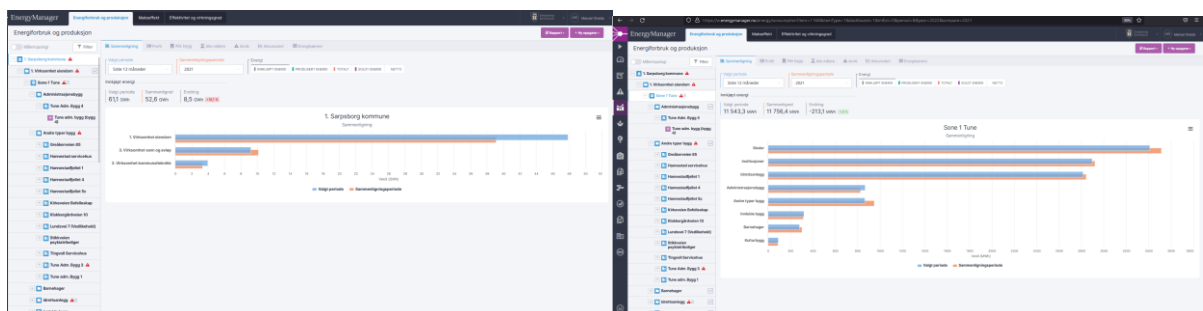
The most important sections of this EMS will be reviewed below:

6.6.2.1. Energy Follow-Up

This is the section where the Energy Manager will obtain the most important information. It is divided in three subsections: Energy Usage and production, Maximum Power, and effectivity and efficiency. It is possible to compare values from different periods and see the changes between them, in a general and small scale.

6.6.2.1.1. Energy Usage and Production

The energy consumed by every building and service is gathered by different devices installed onsite, whose readings are captured through Lhub, which logs readings every minute.



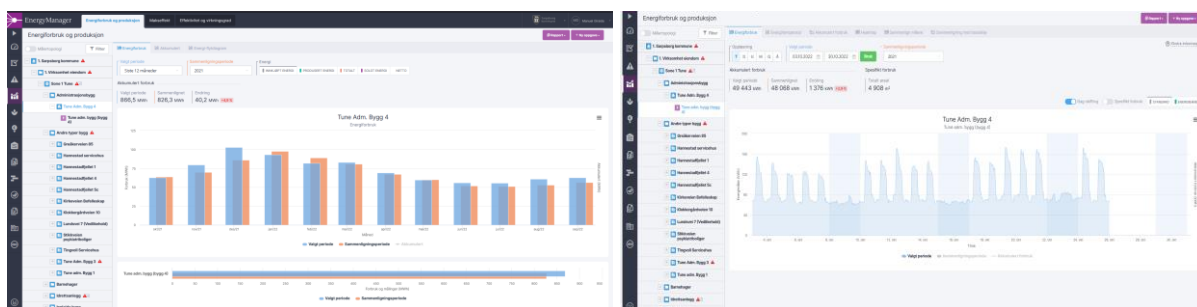


Figure 6-11 Multiple Visualization options for Energy Consumption

Figure 6-11 shows the energy usage charts in different levels of detail: In the upper left there is the summary of the total energy used by the commune during the last 12 months, classified by business line, compared to the total from 2021. In the upper right, a zone was selected, this time it shows the values for zone one, they are classified by type of building, it can be either a school, an institution, an administrative building, etc. In the lower left and lower right, it shows the energy demand of an specific building, but shown in monthly and in hourly values respectively.

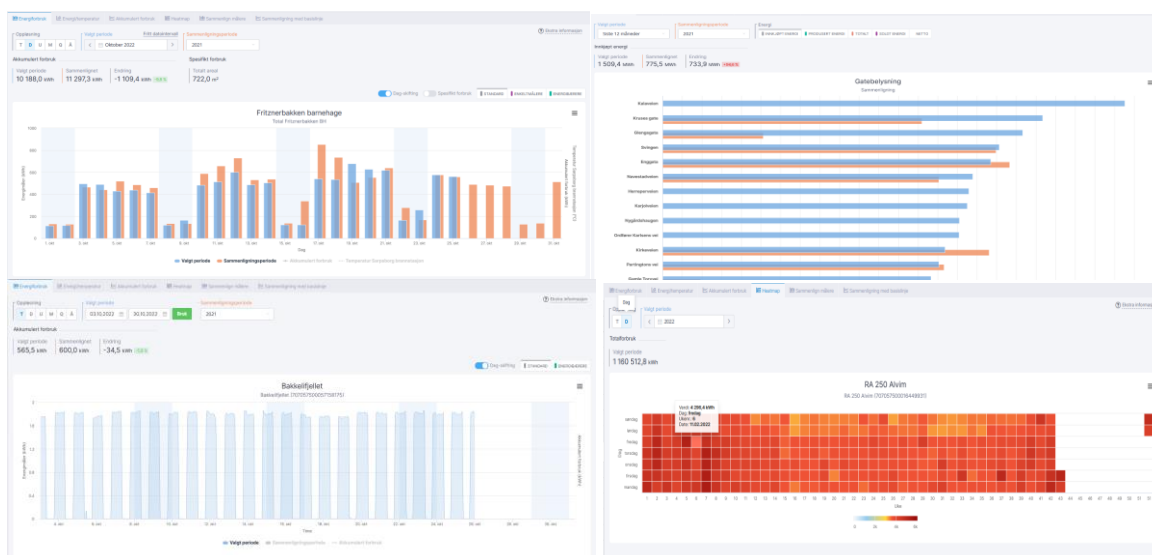


Figure 6-12 Other Energy Demand Visualization Options

Figure 6-12 shows diverse energy usage visualization options. In the top left corner there is the energy consumption of a kindergarten in October of 2022 compared to the values of the same month in 2021. The chart from the top right shows the usage of the street lighting in different zones, whereas the chart in the low-left shows the same but for a specific street during the days of october 2022 (The patterns matches the working pattern of public lighning, the lights are on during the hours when it is dark). On the lower-right a heatmap displays the consumption heatmap of a sewage treatment plant.

The information is also classified by business line, with categories such as: properties, municipal services, or water and drainage. As seen in Figure 6-13.

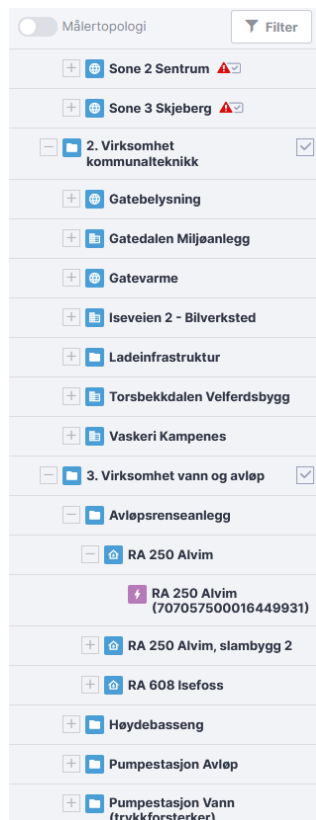


Figure 6-13 Categorization by Business Line

The energy manager can select different types of charts for different types of assets, in a long or short period.

If needed, the user could also see which periods have the most consumption, Figure 6-14 shows a matrix of peak energy demand for a kindergarten. Every column is an hour of the day, whereas every row is a day of the month.

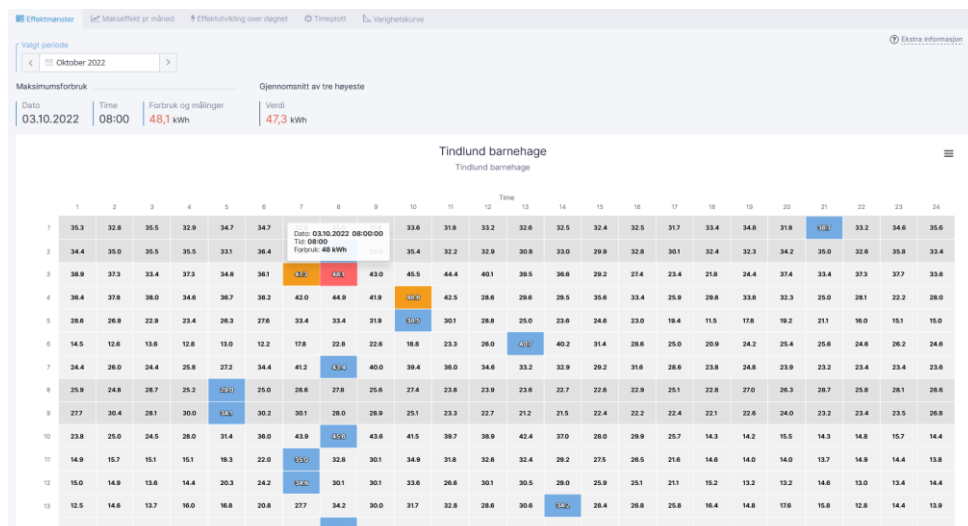


Figure 6-14 Energy Usage Matrix

6.6.2.1.2. Other charts

Figure 6-15 shows the relation between temperature and energy usage. This picture shows only half of the pattern, as the energy demand will increase again for higher temperatures, since the AC will have to be used to cool down a building if the temperature is too high.

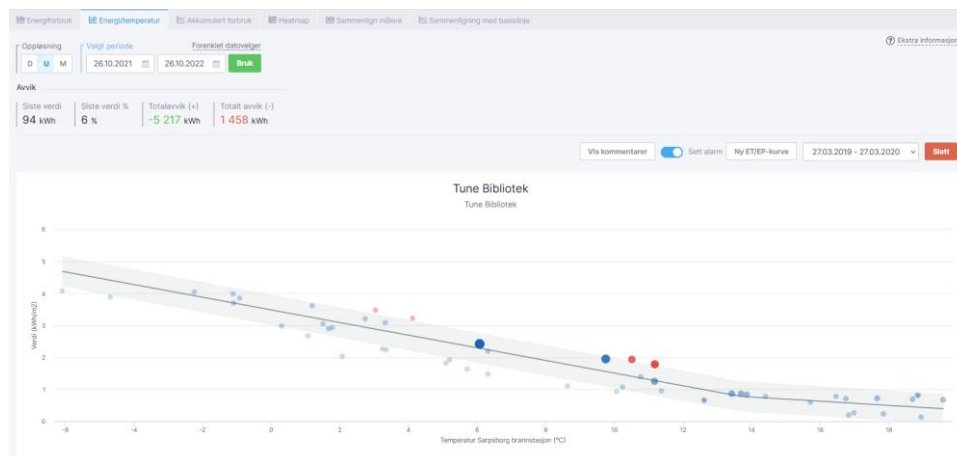


Figure 6-15 Energy Demand vs Temperature

The next picture portrays the historical energy usage of a given asset through the last five years. Although the EMS is monitoring that building, still there is not a considerable improvement in the energy demand for the Tune Bibliotek. Knowing this, the energy manager can further in detail and determine what might be causing this. It is also worth noting the cyclic increments during december, which aligns with the consumption patterns in the north of Europe.



Figure 6-16 Consumption During the Last 5 Years in a Building

A very useful feature is the possibility to view the energy usage pattern of an asset in different scales of time.

Figure 6-17 portrays the hourly behavior of the demand for a Library during the days transurred in October. In this case, the demand peaks between 12:00 and 17:00, which tend to be the times when this type of building is more visited.

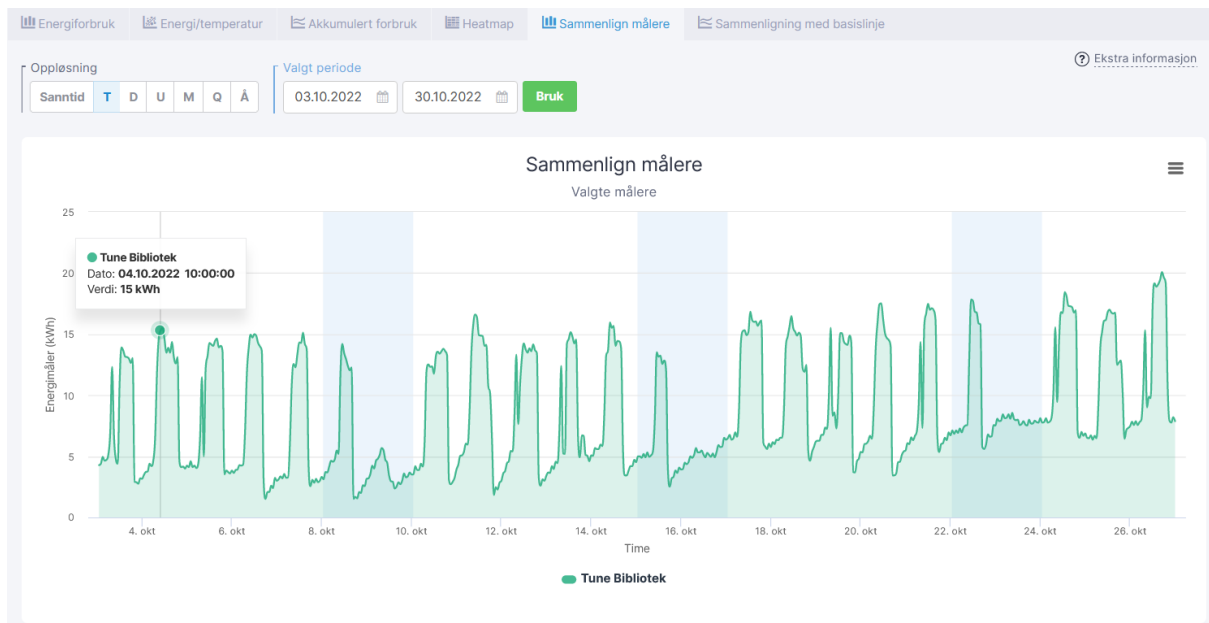


Figure 6-17 Hourly Energy Consumption Pattern

In Figure 6-18 is seen how the demand behaves daily through the month. As expected, the building will use the less amount of energy during the weekends. However, the average values will increase as winter approaches.

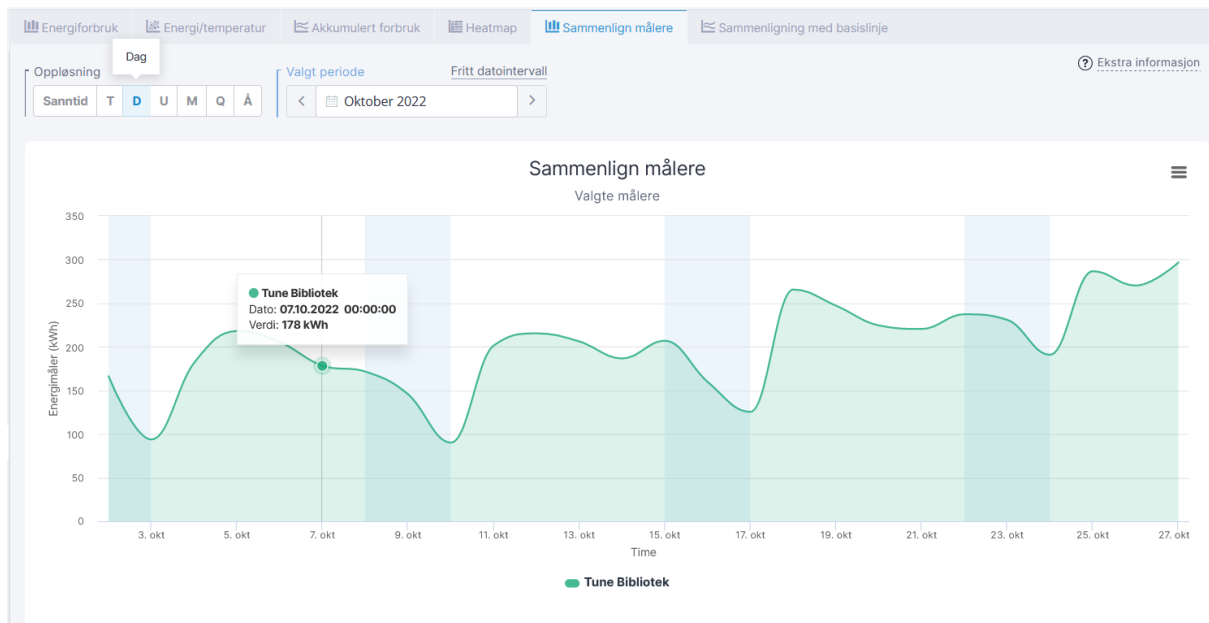


Figure 6-18 Averaged Daily Consumption Pattern

The chart displayed in Figure 6-19 displays the weekly values throughout 2021. In this case, it is seen that the demand drops in the summer months and peaks in the winter months.

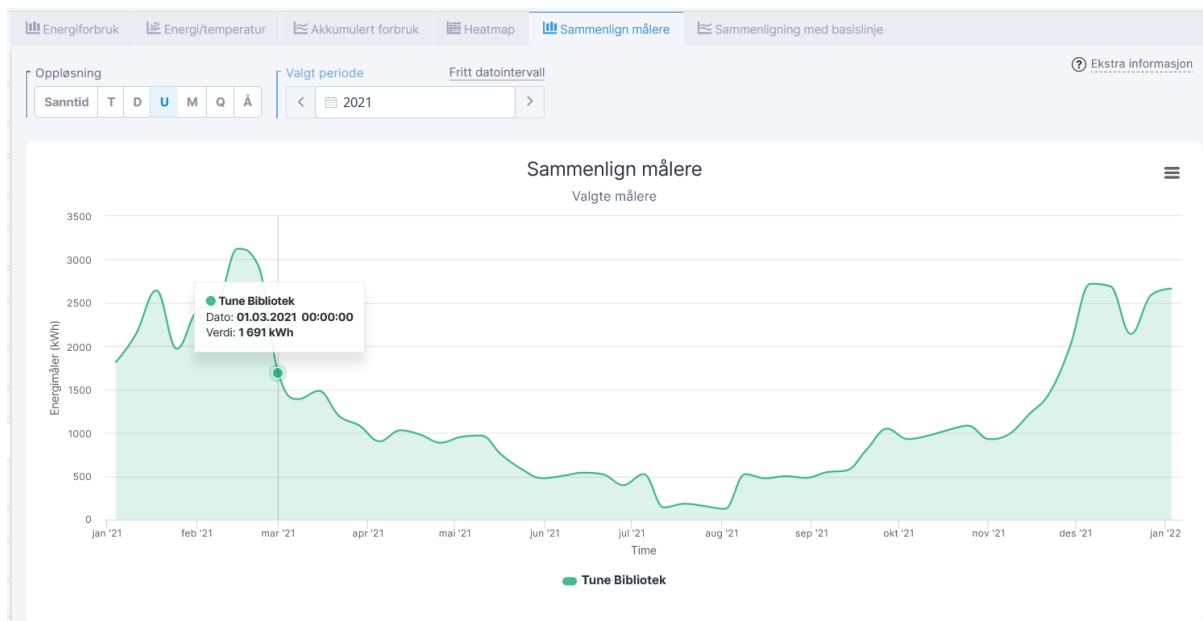


Figure 6-19 Mean Weekly Energy Usage During 2021

Lastly, in Figure 6-20, the quarterly consumption for the last 5 years is shown. In this plot, the cyclic high-low demand pattern is clearly visible.

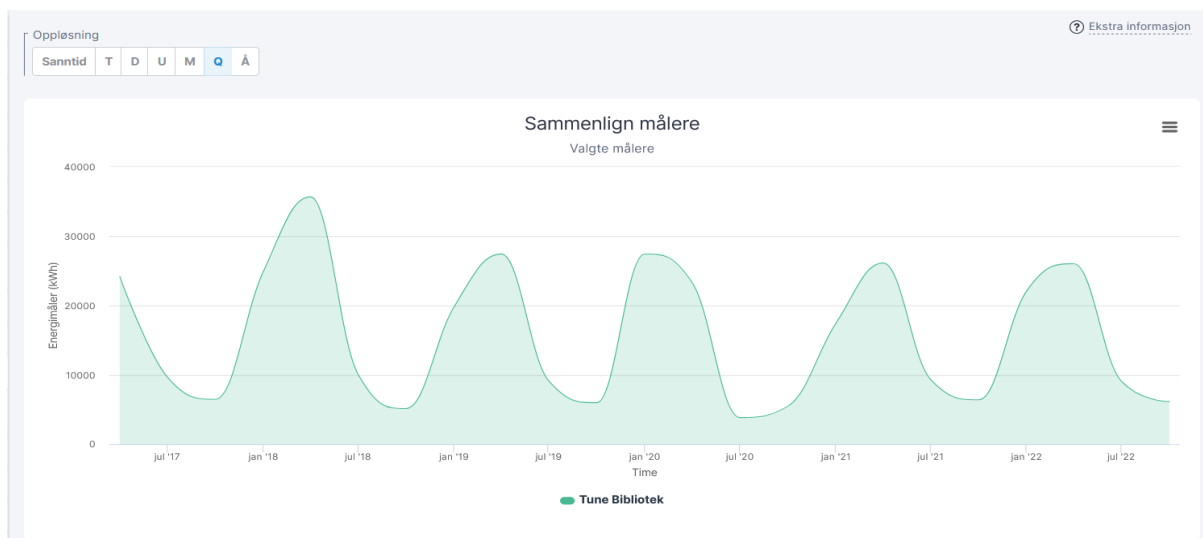


Figure 6-20 Quarterly Energy Demand for the Last Five Years

The energy manager can manipulate and change the scales of all these charts to detect anomalies and act, also to detect opportunities for improvement. For example, they could find that in the weekly chart, the usage does not drop in the weekends, then they could take a look at the library's HVAC and find that it remains on when it is closed. In that case, they could make sure to turn it off and reduce their consumption.

For a more general overview of how effective the measures implemented in the plan and do stages have been, the energy manager can quickly review the chart from Figure 6-21. There they will see the previous and current energy consumption, with the percentual change between

them. This way it will be simple to know If the organization has been successful in achieving their energy goals.

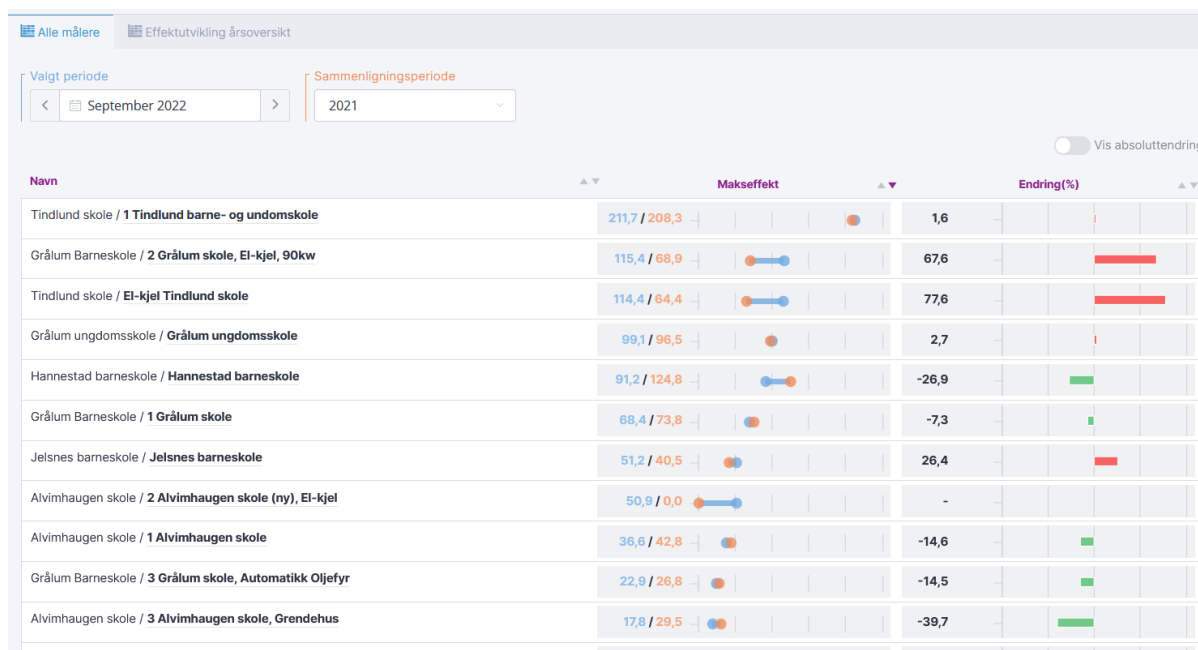


Figure 6-21 General Overview of Changes in Energy Consumption

6.6.2.2. Others

Here, some other sections of the EMS are described

6.6.2.2.1. Alarms

In this section, all the recent alarms are listed. It is possible to see the location where it was activated and its date. Then, it is possible to see in more detail what occurred and what is the status of the alarm. As seen in Figure 6-22, the alarm was acknowledged for the Energy Manager, the target value was 2.56 while the measurement was 3.339. It was activated in October 21st. The cause was a faulty drive belt in a ventilation system.

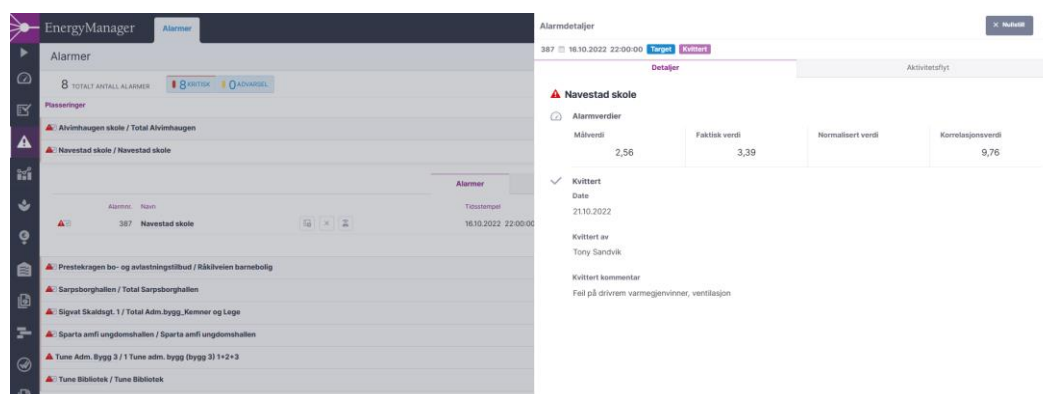


Figure 6-22 Alarms Dashboard

6.6.2.2.2. Environment – CO2 Emissions and Water Consumption

The tool will show the user the total CO2 emissions for a given period, in Figure 6-23, for this case, the estimated emissions for 2022 amount to 572 tons, whereas for 2021 it was 603 tons. The district heating in the sport facilities has a considerable increase, it is likely that the measurements for this category started only on 2022.

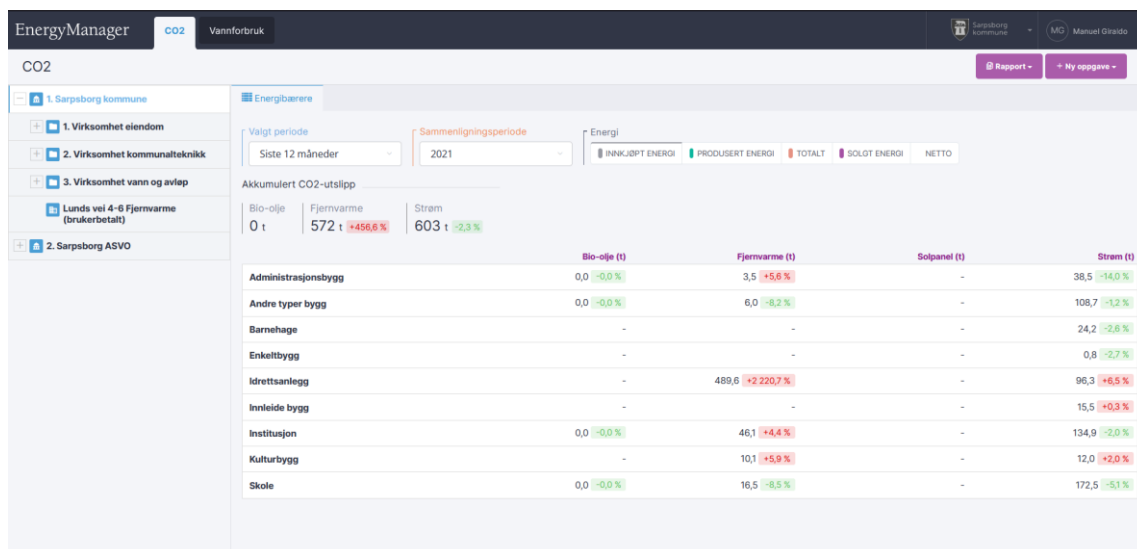


Figure 6-23 CO2 Emission and its Change

6.6.2.2.3. Indoor Climate

With the information about energy consumption, the energy manager now can review the temperature of a given building, the CO2 content, and the air humidity, in real time or during a given month.

Figure 6-24 shows the real time values for air humidity, temperature and CO2 content in conjunction with movement detection inside the building.



Figure 6-24 Real-Time Indoor Climate Values

Figure 6-25 portrays the monthly values for the temperature sensor and the associated heatmap for movement detection.

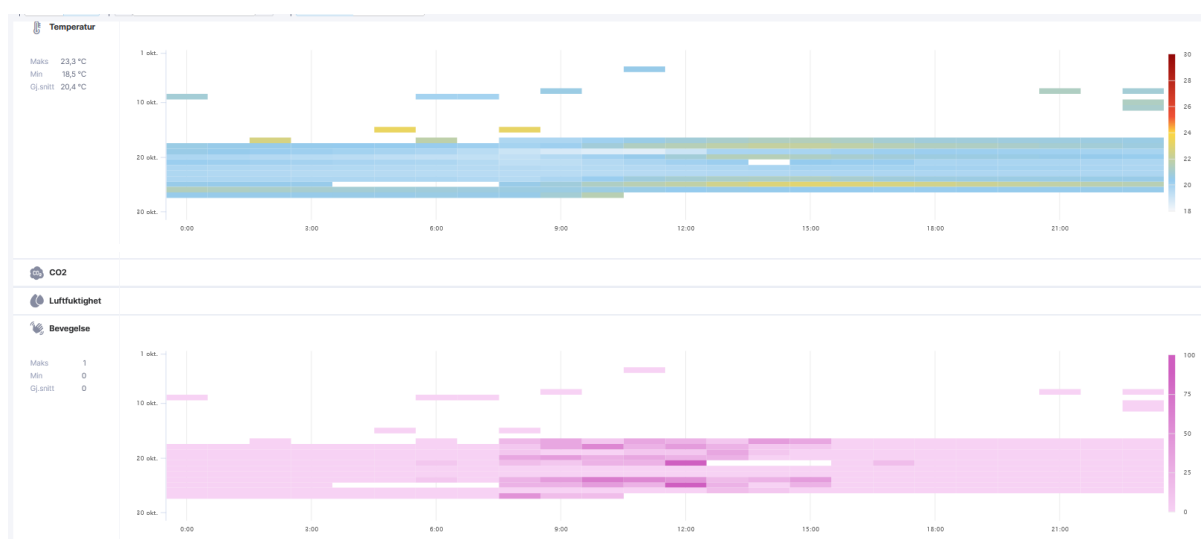


Figure 6-25 Monthly Indoor Climate Heatmap associated with Movement Detection

With these graphs, the energy manager can determine if the ventilation system is working properly and keeping up to the demand from the users, which is calculated given the movement detection values.

6.6.2.2.4. ISO

In this part of the EMS, the organization can describe the aspects that are more related to it in the frame of ISO 50001. As seen in Figure 6-26, the quality team can add directives in a simple manner. This information is then available for the personnel in charge of working with the EMS.

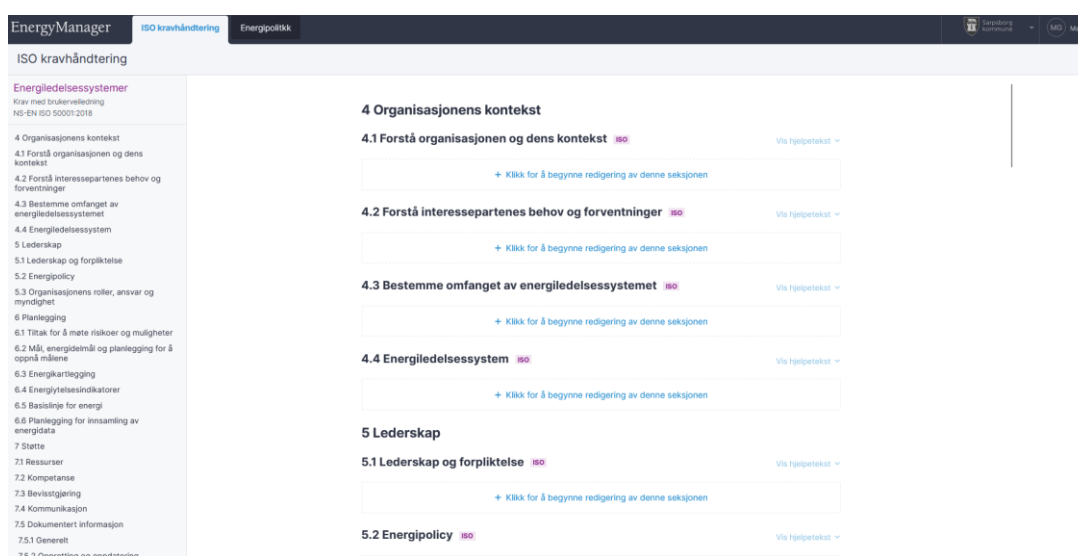


Figure 6-26 ISO 50001 Interface

7. EMS Evaluation and Enova Benchmark

In total, 13 EMS were analyzed to different depths. This depended on how easy was to get in touch with a product or a demo, as it was possible with Energy Manager AS, Siemens, FENECON EMS and Home Assistant. Whereas, with systems such as Ant solutions, and Clevair, it was difficult to get access to a somewhat working product. Solutions like Homey pro and Strømkontroll depended on the existence of devices that needed to be purchased, which in turn limited the level of detail of the analysis for their product.

For the systems that did not have a demo available, the information was gathered from their case studies and websites. It is worth noting that several attempts were to get access to Yodiwo's demo, however, this was to no avail. Therefore, the evaluation of their product is done based on the information provided by them during the workshop from September 9th.

7.1. Enova Benchmark

Enova is an enterprise owned by the Norwegian government, whose purpose is to promote and incentive the responsible and efficient use of energy in Norway. As part of their program, they provide financing for homes and buildings that will implement software solutions to track their energy consumption.

Enova have defined some features that are necessary for a quality EMS and condensed these in a benchmark table. This benchmark will be used to compare the different EMS Systems that were summarized in the previous section. The list is extensive, and the described EMS are very diverse, so the most important features were selected.[40]

7.2. EMS Evaluation

This comparison is purely qualitative and does not represent a ranking in regards of which solution is the best, as these products have different target customers and applicability. The results were gathered in a table which is in Appendix B.

The features are divided in five categories: General, Data Collection, Processing and Analysis, User Interface and Reporting, and Capacity and Security.

7.2.1. General

Except for Energy Lens, all of the products provide a web-based interface, which makes their product more versatile, as it does not depend on an specific operative system to be used. This means also that any of these products can be used in either a PC or a mobile device, given that the front-end development has considered responsive technologies and is compatible with recent web browser technologies.

Not all products show clearly that they have additional modules for water, waste, climate, environment, etc. There is a varied level of detail on what they include. Solutions like Energy Management provide information about water consumption and Co2 emissions. Whereas consumer-oriented EMS are very flexible, it is possible to register information about any smart sensor, whether of temperature or water consumption.

One very important feature that all of the systems lack is clarity about the data ownership. This should be clearly stated, and all the technology providers should ensure that all gathered data belongs to the user.

7.2.2. Data Collection

All the providers have compatibility with different sensors and protocols. Zigbee and Zscale are present mostly in the consumer-oriented software, whereas in corporative solutions there is reliance in protocols like Modbus. Products like Strømkontroll and Energy Manager AS use HAN and Elhub respectively. Almost all the solutions have alternatives to import and export data, whether through json and/or CSV files.

The focus in IoT is generalized, it is important for service providers such as Clevair, that goes to existing buildings and install sensors to obtain a better picture of the energy review. It is expected to see more implementations with IoT, as this greatly increases the flexibility of the system.

7.2.3. Processing and Analysis

In essence, an EMS helps the energy manager predict and take actions about the future energy consumption. So, it is no wonder that the possibility to analyze trends over different time intervals is accomplished by all the reviewed products.

The benchmark relating to the possibility to create alarms and the possibility to express the readings in economical terms is also fulfilled by most of the EMS.

On the other hand, it is not clear that it is possible to correct errors in the data for. None of the reviewed systems gives the option to manually change a reading or to delete data points. This is understandable, as the modification and deletion of information should be made by personnel with the appropriate authorizations. Nevertheless, a good system should have the option to detect and eliminate outliers if necessary.

Other than Energy Manager AS, it is not clear which solutions can provide statistics regarding CO₂, NO_x and SO₂ emissions. This should be an improvement topic for the existing products.

7.2.4. User Interface and Reporting

There is room for improvement in the user support benchmark. Few of the products have user's forum where to ask for help and guidance, or to inform about bugs. The exceptions to this are FENECON EMS, which developed their software with an open-source technology called OpenEMS, and Home Assist, which is on its side an open-source alternative.

7.2.5. Capacity and Security

There is not much information about how the providers handle and back-up the customer's data. This is critical and a point for improvement in all the solutions. The historical data is paramount to update the models and improve the response of the EMS to different scenarios. It is necessary to guarantee that cybersecurity best practices are being followed.

Likewise, it is difficult to say that any of the alternatives has functionalities to administer the users' authorization levels. This also impacts on the cybersecurity aspect, as a poorly

experienced user with an adequate authorization level can delete information by accident, or can act unwittingly as a facilitator for a cyberattack.

As most of the reviewed EMS are web-based, it is possible to scale them as the number of users and services increase.

8. Conclusion

The purpose of the energy management system is to reduce energy consumption and undesirable environmental effects. It is necessary to present the building's energy performance in an adequate and clear way and to use numerical indicators that are based on annual averages and national or regional standards. The company must first have a strategy to define the essential energy uses. The next step is to optimize resources to decrease energy demand and, as a result, lead to cost savings. Subsequently controlling the process is crucial, after that, the system's performance must be assessed for systematic improvement. Finally, the possibilities for enhancing energy performance should be investigated, therefore, the company will be able to realize the requirements for changing the energy policy. The reduction of demand for energy consumption can be used as evidence of improvement in energy performance.

The evaluated EMS cover different target customers: Industry, corporate buildings, and homes. Almost all of them are aligned with ISO50001. These systems should provide the user to view their energy statistics in different scales of time and size, this will help them obtain more insight from the information displayed and see the energy consumption patterns in all these scales. In turn, this will enable the user to better identify scenarios that are outside of the normal behavior and to make use of improvement opportunities in a timely manner.

So far, there is not a product that offers machine learning integration, however, as this is a data rich field, is safe to think that this will change soon. As the literature review showed, a good EMS accounts for all this data and uses it to optimize the control response of the equipment inside a building. Many of the solutions allow the user to control on-off responses in the equipment, but there is not much information about the use of technologies like modern predictive control besides products like Sentron from Siemens, which is almost solely intended to be integrated with their own brand of hardware products. This would make integration with other EMS difficult.

Nevertheless, the main goal of these products currently is providing enough information to the energy manager to make decisions with better criteria, not only in terms of reduce the energy consumption, but to help their organization operate a viable business. Therefore, although many EMS still do not use sophisticated control algorithms, the information provided by them helps their users obtain savings of up to 40% by improving their infrastructure or by changing their consumption patterns, which could be described as actions that do not require automation or control.

There is an increasing proportion of smart sensors and devices. It is expected that this tendency is maintained since the use of these provides the EMS with increasing flexibility. The Energy Manager would not need to incur in expensive installations to obtain more insight in a space inside a building. It could be a matter of having Wi-Fi and sensors and actuators that can be accessed through the internet. As expected, this poses cybersecurity challenges that need to be addressed, EMS providers have an opportunity to be clearer about how the information is stored, managed and backed-up.

There should be a wide interest in using open-source technologies for EMS development, also, each EMS software should include an forum where the Energy Managers can discuss and obtain help. An Energy Manager is not a simple user that reviews the charts, this person should also be able to develop scripts to implement changes and control the hardware devices under their supervision.

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Appendices

Appendix A

Energy management system questionnaire

1. There is a list that shows the energy management system's advantages:

- (a) Reduce administrative expenses
- ✓ (b) Decrease energy costs
- (c) Verify energy conservation measures
- (d) Describe variations in the energy budget
- (e) Advanced and modern analytics
- (f) Greenhouse gas reductions

Please select the items which are important for your company.

2. There is a list of energy management attributes:

- ✓ (a) A user-friendly design with charts and visual information
- (b) Specific data based on your needs
- (c) Predicting processes and evaluating the possible effects of energy efficiency

Please select the items which are necessary if you implement an energy management system in your company

3. The following modules should be included in the approach for calculating the energy performances of buildings:

- (a) The building's thermal characteristics (shell and internal partitions, etc.)
- ✓ (b) Heating installation and hot water supply, including their insulation characteristics
- (c) Air-conditioning installation
- ✓ (d) Ventilation
- (e) Built-in lighting installation (mainly in the non-residential sector)
- (f) Building placement and orientation, as well as external climate
- (g) Passive solar systems and solar protection
- (h) Natural ventilation
- (i) Indoor climatic factors, including the designed indoor climate

what data is available? Are you willing to provide the data to implement an energy management system in your company?

4. Energy management platform solutions typically differ in terms of:

- (a) Visualization and reporting
- (b) Fault detection and diagnostics
- (c) Predictive maintenance
- ✓ (d) Continuous improvement, and optimization

which one is operated by your company or its crucial for your company?

5. Passive buildings, which require so very little energy that they don't require a heating or cooling system, are in many ways a desirable alternative to typical buildings, but they face numerous obstacles. Listed here are several challenges:

- ✓ (a) Financial barriers
- (b) Inadequate understanding of the importance of energy efficiency
- (c) Lack of support by governments or other public authorities
- ✓ (d) Lack of positive examples for the citizens and the companies

which one is the most important barrier to developing passive building?

Appendix A <Appendix title>x

Appendix B EMS Comparison Results

Appendix C <Appendix title> etc.

Remark: If new appendix ➔ New page (i.e. don't put several appendices on the same page)